

CAPM Exam Prep Seminar

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CAPM requirements

- Education requirements
 - High school diploma
 - 23 contact hours needed
- Certification maintenance requirements
 - 15 PDUs per three-year cycle to maintain certification
- Exam information
 - 3 hours, 150 questions
 - 15 pre-test questions

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CAPM exam information

- Fundamentals of project management and the role of project managers
- Project management environment, and project integration management
- Project scope, schedule, cost, quality and resource management
- Project procurement, communication, stakeholder and risk management

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CAPM and PMBOK Guide

- All exam questions come from at least two references
- PMBOK® Guide – Seventh Edition
- PMIstandards+™
- The PMI Guide to Business Analysis (2017)
- Business Analysis for Practitioners: A Practice Guide (2015)
- Agile Practice Guide (2017)
- The Project Management Answer Book
- Effective Project Management: Traditional, Agile, Extreme, Hybrid (8th Edition).

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CAPM Exam Content Outline

- Project Management Fundamentals and Core Concepts: 36 percent
- Predictive, Plan-Based Methodologies: 17 percent
- Agile Frameworks/Methodologies: 20 percent
- Business Analysis Frameworks: 27 percent

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Walk-through CAPM Exam Content Outline

- Essential to understand the ECO
- ECO is what's on your exam
- Don't skip studying the ECO

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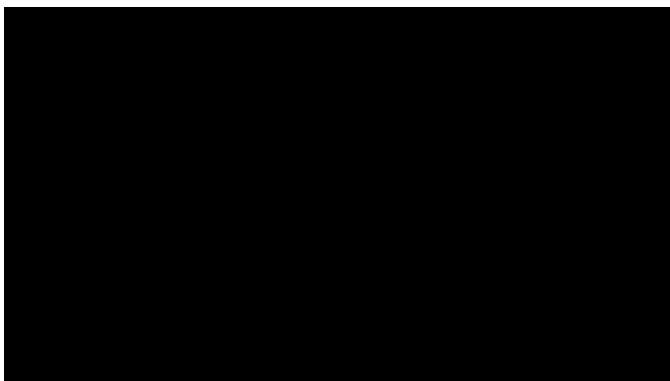


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Section overview

Demonstrate an understanding of the various project life cycles and processes

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Distinguish between a project, program and a portfolio

10

What is a project?

- Temporary effort
- Goal and deliverables
- Start and end date
- Clear scope and objectives
- Specific resources and budgets

11

Project examples:

- Building a new bridge
- Developing a new software application
- Renovating a house

12

What is a program?

- Group of related projects that work towards a common objective
- Multiple projects with interdependencies
- Larger scope than a single project
- Longer timeline

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Program examples:

- A government infrastructure development program
- A company's employee training program
- A research and development program

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What is a portfolio?

- Collection of projects, programs, and operations that align with strategic objectives
- Broad scope that encompasses all work within an organization
- Used to manage and prioritize projects and programs

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Program examples

- Company's product portfolio
- Government agency's infrastructure portfolio
- Skyscraper construction

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Summation of portfolio, program, projects

- Project: temporary, specific goal, deliverables
- Program: multiple projects, common objective, larger scope
- Portfolio: collection of work, strategic alignment, broad scope

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Advantages of project, program, and portfolio

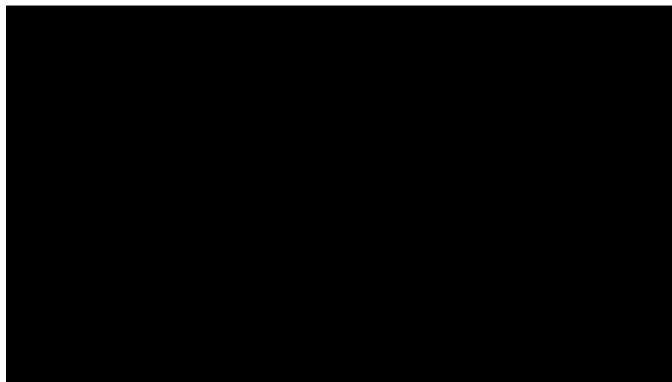
- Project: agile, responsive to change, focused on specific goals
- Program: efficient, cost-effective, coordination of multiple projects
- Portfolio: strategic, alignment with long-term goals, broad scope

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Disadvantages of project, program, and portfolio :

- Project: risky, potential for failure, limited scope
- Program: complex, coordination challenges, longer timeline
- Portfolio: difficulty in managing and aligning with changing priorities, broad scope

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Distinguish between a project and operations

21

Projects

- Temporary endeavor
- Specific goal or objective
- Start and end date
- Defined budget

22

Operations

- Ongoing work of an organization
- Produce goods or services
- Accounting, IT sales
- Lines of business

23

Projects and operations

- Projects have a defined budget
- Operations have ongoing costs
- Operations involve ongoing production
- Project managers
- Operational managers

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Distinguish between predictive and adaptive approaches

26

Predictive projects

- Assumption that the future can be accurately predicted based on past trends and data
- Involves long-term planning to predict future events or outcomes
- Detailed plans and budgets
- Allows for more accurate forecasting and planning
- May be less flexible and less able to adapt to unexpected changes

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Adaptive projects

- Assumption that the future is uncertain and subject to change
- Flexible planning and decision-making
- Ability to adapt to changing circumstances
- Short-term planning and focus on quick adaptation
- More flexible and able to adapt to unexpected changes
- May be less efficient and more costly due to frequent adjustments

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Predictive projects

- Planning and decision-making
- Assumption that the future can be accurately predicted based on past trends and data
- Construction
- Financial planning
- Industrial work

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Adaptive projects

- Assumption that the future is uncertain and subject to change
- Plans and decisions should be flexible and adapt to changing circumstances
- Software development
- Crisis management

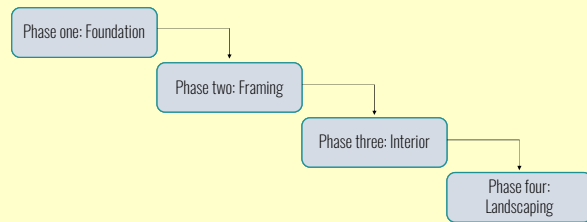
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Differences: predictive and adaptive projects

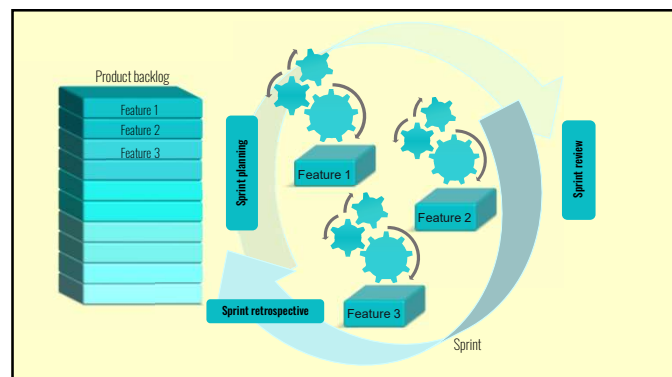
- Assumption of predictability
- Assumption of uncertainty
- Focus on long-term planning
- Focus on short-term adaptability

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Predictive approach



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Predictive project examples

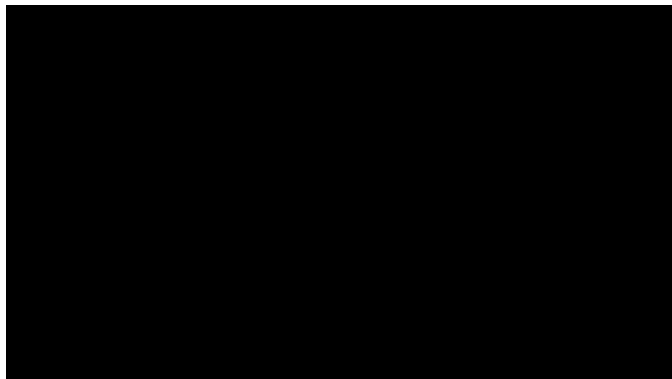
- Financial planning: create a long-term financial plan based on past performance and trends
- Construction: plan and budget based on past data about construction costs, materials, and labor
- Manufacturing: plan production schedules and inventory levels based on past demand data
- Weather: forecast based on past data about weather patterns and atmospheric conditions

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Adaptive project examples

- Software dev: build and test software, adjusting based on changing user needs and feedback
- Marketing: plan and execute a campaign, adjusting strategy and tactics based on feedback
- Crisis management: respond to unexpected events or emergencies
- Events: large-scale event, adapting to changes in attendance, weather, or other factors

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Project management life cycle

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Universal to all projects

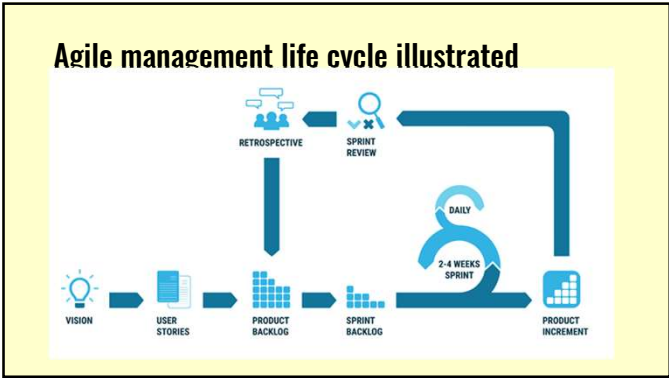
- Initiating
- Planning
- Executing
- Monitoring and controlling
- Closing

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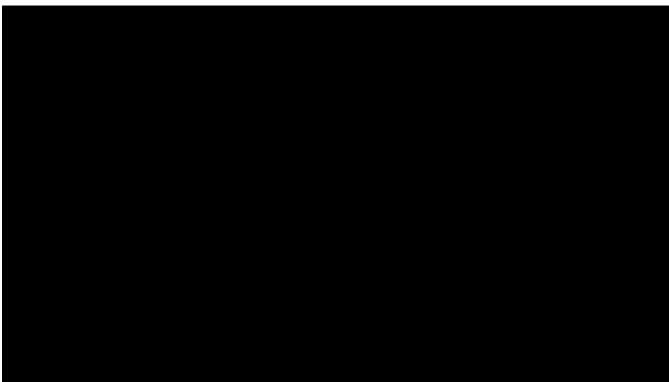
Project management life cycle illustrated



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Distinguish between issues, risks, assumptions, and constraints

42

Project risk

- Uncertain event or condition
- Positive or negative effect
- Scales of probability and impact
- Identified, analyzed, documented, response
- Risk register

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Pure risk and business risk

- Pure risk – loss of life or limb, dangerous, always negative
 - Safety measures
 - Electrical work and licensing
- Business risk – loss or return of investment; positive or negative
 - Return on investment
 - Stock market
 - Time or cost savings

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Examples of risk

- Technical failure
- Data security
- Dependency on external resources
- Technical risk
- Schedule risk
- Budget risk
- Resource risk

45

Project issues

- Risk events that have occurred
- Issue log
- Issue owner
- Issue resolution date
- Technical failure
- Data security
- Dependency on external resources

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Assumptions

- Believe to be true
- Not proven to be true
- Assumptions may be risk

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Examples of assumptions

- Availability of necessary resources
- Stakeholder support
- Technical feasibility
- Legal and regulatory compliance

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Project constraints

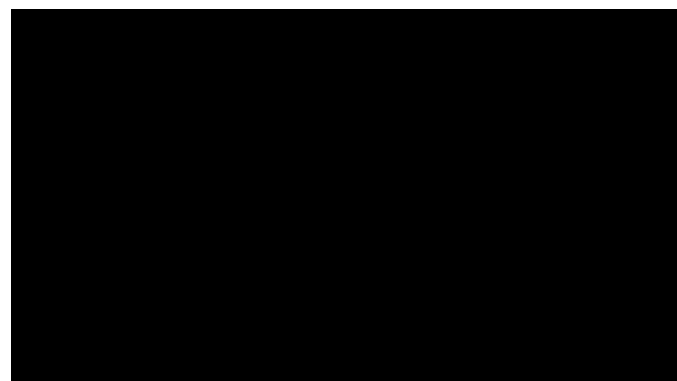
- Anything that limits your options
- Requirements for the project

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Examples of project constraints

- Budget
- Deadline
- Must finish on /must start on (MFO/MSO)
- Resources
- Vendors
- Requirements

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Review/critique project scope

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Defining the project scope

- Defines the boundaries of a project
- Outlines what will be included or excluded
- Detailed description of objectives, deliverables, and requirements

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Project scope statement

- Goals and objectives
- Deliverables
- Requirements
- Constraints
- Assumptions
- Exclusions
- High-level risks

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Review/critique project scope

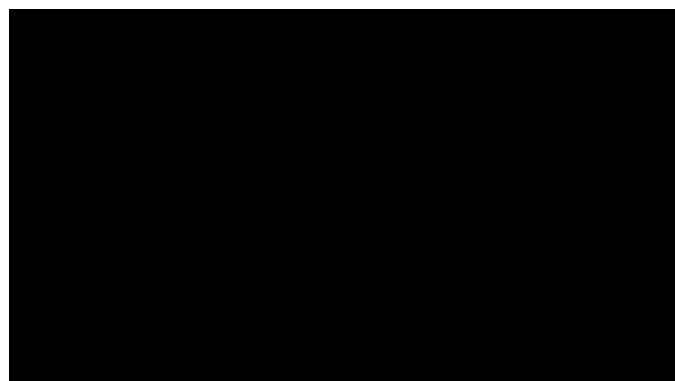
- Are the objectives defined and achievable within the project scope?
- Do objectives align with the overall goals of the organization?
- Are the deliverables realistic and achievable?
- Are requirements necessary and relevant?

55

Review/critique project scope

- Are constraints reasonable and necessary?
- Are constraints defined and understood by the project team?
- Are assumptions realistic and reasonable?
- Are exclusions reasonable to keep the project within scope?

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49 project management processes

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PMI standards™

- Five process groups: IPBCC
- Ten knowledge areas
- Intersection of KA and PG is a process

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Project integration management

- Developing project charter
- Developing project management plan
- Directing and managing project work
- Monitoring and controlling project work
- Performing integrated change control
- Closing the project

60

Project scope management

- Planning scope management
- Collecting requirements
- Defining scope
- Creating the WBS
- Validating scope
- Controlling scope

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Project schedule management

- Planning schedule management
- Defining project activities
- Sequencing activities
- Estimating resources and duration
- Developing a schedule
- Controlling the schedule

62

Project cost management

- Planning cost management
- Estimating costs
- Determining a budget
- Controlling costs

63

Project quality management

- Planning quality
- Managing quality
- Controlling quality

64

Project resource management

- Planning resource management
- Estimating activity resources
- Acquiring resources
- Developing the team
- Managing the team
- Controlling the resources

65

Project communications management

- Planning communications management
- Managing communications
- Monitoring communications

66

Project risk management

- Planning risk management
- Identifying risks
- Performing qualitative risk analysis
- Performing quantitative risk analysis
- Planning risk responses
- Implementing risk responses
- Monitoring risk

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Project procurement management

- Planning procurement management
- Conducting procurement
- Controlling procurement
- Closing procurement

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Project stakeholder management

- Identifying stakeholders
- Planning stakeholder management
- Managing stakeholder engagement
- Controlling stakeholder management

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Apply the project management code of ethics to scenarios

71

PMI Code of Ethics

- Must agree for exam and credential
- Download from PMI.org

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Vision and applicability

- Committed to doing what is right and honorable
- High standards for themselves and strive to meet these standards
- PMI code of ethics and professional conduct outlines expectations
- Instill confidence in the profession

73

Code of ethics and professional conduct

- All PMI members
- Non-members who hold a PMI certification
- Non-members who apply to commence a PMI certification process
- Non-members who serve PMI in a volunteer capacity

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Aspirational and mandatory conduct

- Aspirational and mandatory standards
- Aspirational standards: should strive to uphold
- Mandatory standards: requirements: may limit or prohibit behavior
- Practitioners who do not adhere to the mandatory standards may be subject to disciplinary procedures

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Responsibility: aspirational standards

- Make decisions and take actions based on the best interests of society, public safety, and the environment
- Accept only assignments that are consistent with our background, experience, skills, and qualifications
- Only bid on work that our organization is qualified to perform and assign only qualified individuals to perform the work
- We fulfill the commitments that we undertake – we do what we say we will do
- When we make errors or omissions, we take ownership and make corrections promptly

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Responsibility: aspirational standards

- When we discover errors or omissions caused by others, we communicate them to the appropriate body as soon they are discovered
- We accept accountability for any issues resulting from our errors or omissions and any resulting consequences
- We protect proprietary or confidential information that has been entrusted to us
- We uphold this code and hold each other accountable to it

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Responsibility: mandatory standards

- We inform ourselves and uphold the policies, rules, regulations and laws that govern our work, professional, and volunteer activities
- We report unethical or illegal conduct to appropriate management and, if necessary, to those affected by the conduct
- We bring violations of this code to the attention of the appropriate body for resolution
- We only file ethics complaints when they are substantiated by facts
- We pursue disciplinary action against an individual who retaliates against a person raising ethics concerns

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Respect: aspirational standards

- We inform ourselves about the norms and customs of others and avoid engaging in behaviors they might consider disrespectful
- We listen to others' points of view, seeking to understand them
- We approach directly those persons with whom we have a conflict or disagreement
- We conduct ourselves in a professional manner, even when it is not reciprocated

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Respect: mandatory standards

- We negotiate in good faith
- We do not exercise the power of our expertise or position to influence the decisions or actions of others in order to benefit personally at their expense
- We do not act in an abusive manner toward others
- We respect the property rights of others

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Fairness: aspirational standards

- We demonstrate transparency in our decision-making process
- We constantly reexamine our impartiality and objectivity, taking corrective action as appropriate
- We provide equal access to information to those who are authorized to have that information
- We make opportunities equally available to qualified candidates

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Fairness: mandatory standards

- Conflict of interest:
- We proactively and fully disclose any real or potential conflicts of interest to the appropriate stakeholders
- When we realize that we have a real or potential conflict of interest, we refrain from engaging in the decision-making process or otherwise attempting to influence outcomes, unless or until: we have made full disclosure to the affected stakeholders; we have an approved mitigation plan; and we have obtained the consent of the stakeholders to proceed

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Fairness: mandatory standards

- Favoritism and discrimination:
- We do not hire or fire, reward or punish, or award or deny contracts based on personal considerations, including but not limited to, favoritism, nepotism, or bribery
- We do not discriminate against others based on, but not limited to, gender, race, age, religion, disability, nationality, or sexual orientation
- We apply the rules of the organization (employer, project management institute, or other group) without favoritism or prejudice

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Honesty: aspirational standards

- We earnestly seek to understand the truth
- We are truthful in our communications and in our conduct
- We provide accurate information in a timely manner
- We make commitments and promises, implied or explicit, in good faith
- We strive to create an environment in which others feel safe to tell the truth

84

Honesty: mandatory standards

- We do not engage in or condone behavior that is designed to deceive others, including but not limited to, making misleading or false statements, stating half-truths, providing information out of context or withholding information that, if known, would render our statements as misleading or incomplete
- We do not engage in dishonest behavior with the intention of personal gain or at the expense of another

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Code of ethics terms

- Abusive manner: conduct that results in physical harm or creates intense feelings of fear, humiliation, manipulation, or exploitation in another person

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Code of ethics terms

- Conflict of interest: a situation that arises when a practitioner of project management is faced with deciding or doing some act that will benefit the practitioner or another person or organization to which the practitioner owes a duty of loyalty and at the same time will harm another person or organization to which the practitioner owes a similar duty of loyalty
- The only way practitioners can resolve conflicting duties is to disclose the conflict to those affected and allow them to make the decision about how the practitioner should proceed

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Code of ethics terms

- Duty of loyalty: a person's responsibility, legal or moral, to promote the best interest of an organization or other person with whom they are affiliated
- Project management institute (PMI): the totality of the project management institute, including its committees, groups, and chartered components such as chapters, colleges, and specific interest groups
- PMI member: a person who has joined the project management institute as a member

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Code of ethics terms

- PMI-sponsored activities: activities that include, but are not limited to, participation on a PMI member advisory group, PMI standard development team, or another PMI working group or committee; this also includes activities engaged in under the auspices of a chartered PMI component organization—whether it is in a leadership role in the component or another type of component educational activity or event

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Code of ethics terms

- Practitioner: a person engaged in an activity that contributes to the management of a project, portfolio, or program, as part of the project management profession
- PMI volunteer: a person who participates in PMI-sponsored activities, whether a member of the Project Management Institute or not

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Explain how a project can be a vehicle for change

92

Projects are about change

- MACD
- Clear goal that addresses the problem or opportunity
- Stakeholder identification
- Consider the potential impact

93

Planning and organizing the project

- Define the scope and objectives of the project
- Create a detailed plan and timeline for completing the project
- Assemble a team or group of volunteers to help with the project
- Identify any potential challenges or obstacles

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Implementing the project

- Adaptive or predictive
- Keep track of progress and make adjustments
- Stakeholder engagement and communication
- Celebrate successes and milestones along the way

95

Evaluating and adjusting the project

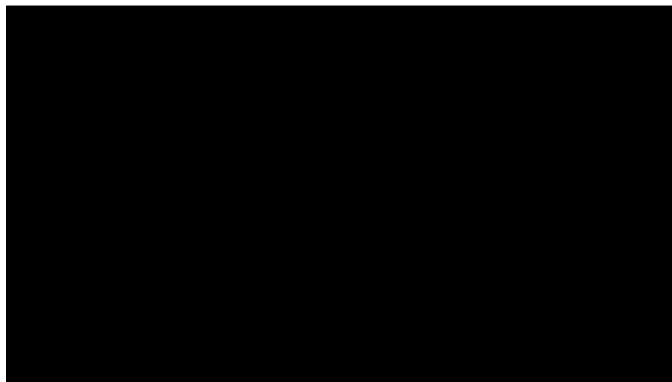
- Key performance indicators
- Milestones and product releases
- Use feedback and data
- Consider long-term sustainability

96

Sharing the results and promoting the project

- Share the results of the project with the community or group
- Share the project with others, including through social media and other outreach efforts
- Lessons learned and reviews

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CAPM Coach

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Section conclusion

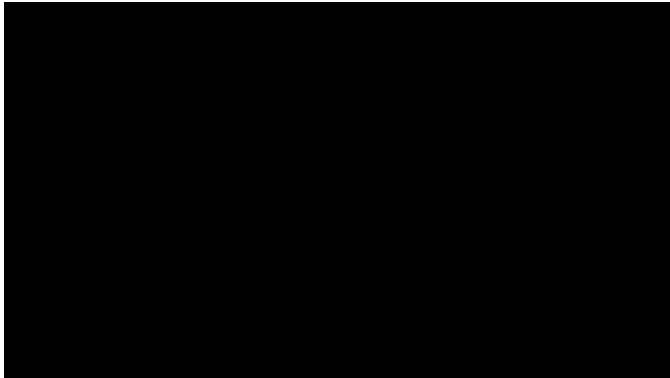
Demonstrate an understanding of the various project life cycles and processes

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Section overview

Demonstrate an understanding of project management planning

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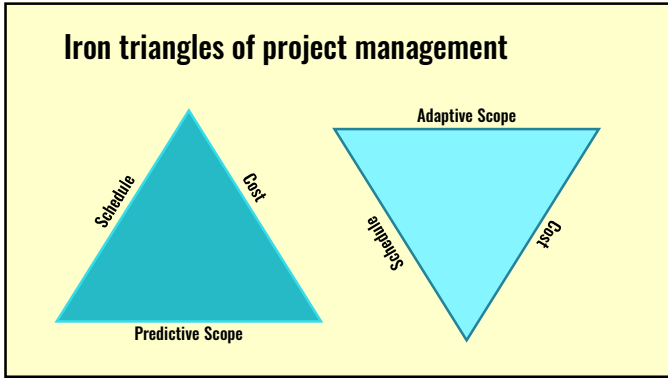
Describe the purpose and importance of cost, quality, risk, schedule, etc.

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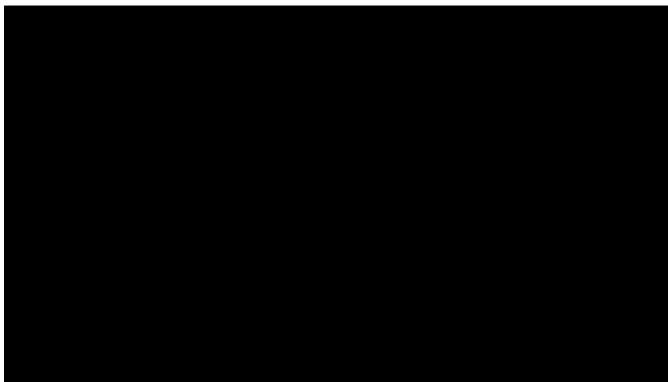
Describe the purpose and importance of cost, quality, risk, schedule, etc.

- Predictive project – predict what will happen
- Adaptive project – adapts to what is happening

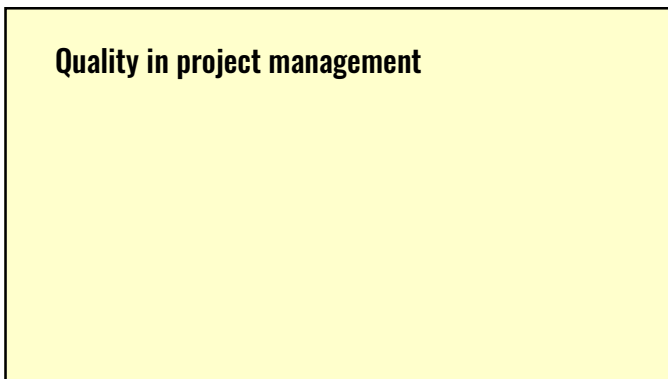
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Quality in project management

- Quality is:
 - Totality of an entity
 - Satisfies needed abilities
 - Stated or implied needs
- Quality processes
 - Quality planning
 - Quality assurance (QA)
 - Quality control (QC)

109

Quality planning

- Planning is an iterative process
- PMI theme: plan, implement, measure, react—and document
- If the product is unacceptable, the project is not finished
- Must meet scope to achieve quality

110

Cost of quality

- Cost of conformance to requirements
 - Safety measures
 - Team development and training
 - Proper materials and processes
- Cost of non-conformance to requirements
 - Liabilities, loss of life or limb
 - Rework/scrap
 - Lost business

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Types of quality costs



PREVENTION

Quality assurance
Deliver the exact project scope and the expected quality
Examples: training, safety measures, and right tools and equipment



APPRAISAL

Quality control
Cost of measuring, testing, auditing, and evaluating
Time for testing



FAILURE

Internal failure:
Scrap, and rework
External failure:
Loss of sales, loss of customers, downtime, and damage to reputation

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Quality management plan

- Quality standards
- Quality objectives
- Roles and responsibilities
- Deliverables and processes subject to quality review
- Quality control and quality management activities
- Quality tools
- Dealing with nonconformance, corrective actions procedures, and continuous improvement procedures

113

Quality metrics

- Quality metrics describe a project or product characteristic
- Control Quality to verify compliance
 - Percentage of tasks completed on time
 - Cost performance
 - Failure rate
 - Number of defects identified per day
 - Total downtime per month
 - Errors found per line of code
 - Customer satisfaction scores

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Manage quality

- Executing process
- Sum of the planning and the implementations
- Sometimes referred to as quality assurance (QA)
- QA should occur before and during the project
- Continuous process improvement
- Managing quality is everyone's responsibility
- QA department could complete the QA activities

115

Completing a quality audit

- Document the best practices used
- Document any variances
- Recommend best practices
- Implement recommendations for quality improvement
- Document quality audit in lessons learned

116

Design for X

- Design for X is a philosophy in product design
- X can mean excellence
- Specific characteristic of a solution

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Design for X

- Design for X is also known as DfX
- X is usually a variable that the project is trying to address
 - Such as cost, uptime, return on investment
 - Considers all components of the design and how the component affects the X variable
 - Goals include lowered costs and improved service, reliability, safety, and overall quality

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Implementing problem-solving techniques



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Results of managing quality

- Quality reports
- Test and evaluation documentation
- Change requests

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Results of managing quality

- Project management plan updates
 - Quality management plan
 - Scope baseline
 - Schedule baseline
 - Cost baseline
- Project document updates
 - Issue log
 - Lessons learned register
 - Risk register

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Controlling quality in a project

- Monitor and measure project results
- Root-cause analysis follows the quality control
- Root-cause analysis helps determine the cause
- Apply corrective actions
- QC occurs throughout the life of a project

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Controlling quality

- Product and the project
- Project management processes
- Measures performance, scheduling, and cost
- Management of the project should be of quality

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Controlling quality themes

- Conduct statistical QC, such as sampling and probability
- Inspect the product to keep errors away from the customer
- Perform attribute sampling and variable sampling
 - Attribute sampling: defect or not?
 - Variable sampling: how defective? How far from quality acceptance?

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Controlling quality themes

- Study special causes to determine anomalies to quality
- Research random causes to determine expected variances
- Check the tolerance range
- Observe control limits

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Inspecting results

- Reviews
- Product reviews
- Audits
- Walkthroughs

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Data gathering and analysis

- Checklists: ensure the work is done the same way each time
- Check sheets:
 - Help organize data about a quality issue
 - Also known as tally sheets
 - Can be used to "tally" up the type of defects

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Testing and product evaluations

- Testing: tests the product against quality standards
- Unique to the discipline in which the project takes place
 - Software development aims to find bugs and errors
 - Construction confirms electrical, plumbing, and HVAC systems
 - Done throughout the project, not just at the end

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Statistical sampling

- Randomly select a few items from the pool of deliverables
- Measure quality on these items as a representative of the whole
 - Installing 1,000 doors
 - Select 200 doors to inspect

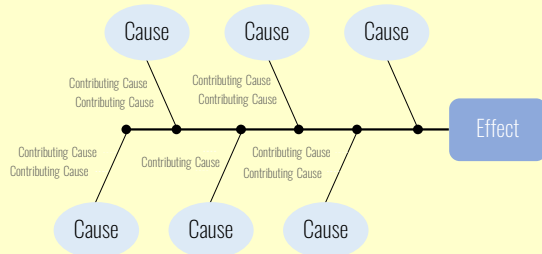
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Cause-and-effect charts

- Show the relationships between the variables within a process
- How those relationships may contribute to inadequate quality
- Can help organize both the process and team opinions
- To create a diagram:
 - Start with the effect
 - Fill out the diagram with possible causes
- Also known as Ishikawa diagrams and fishbone diagrams

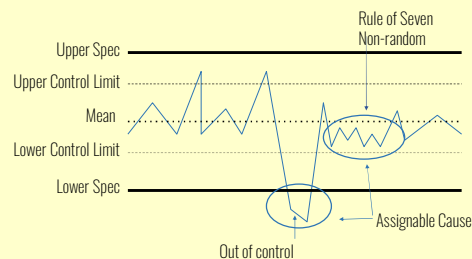
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Cause-and-effect charts

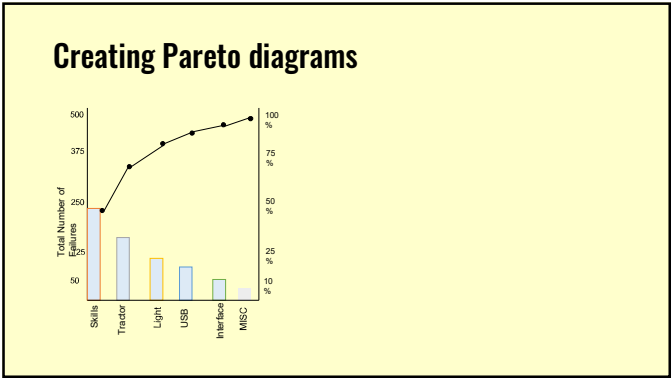


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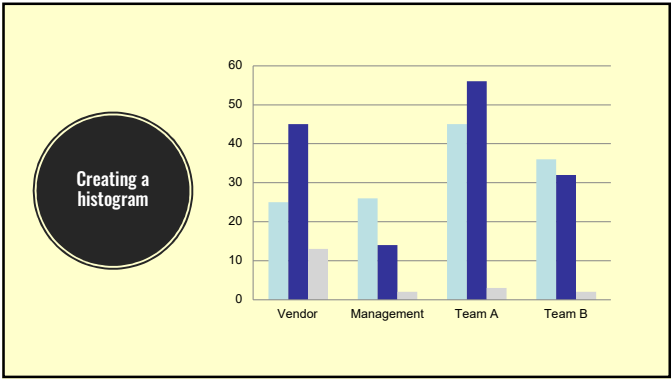
Examining a control chart



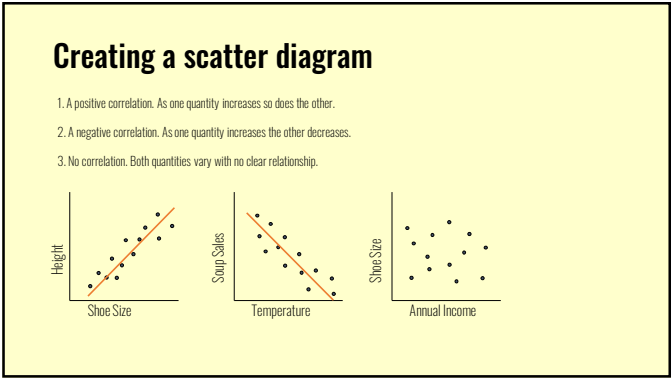
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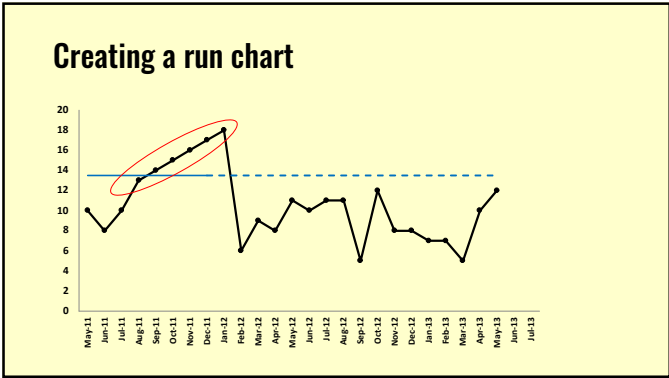
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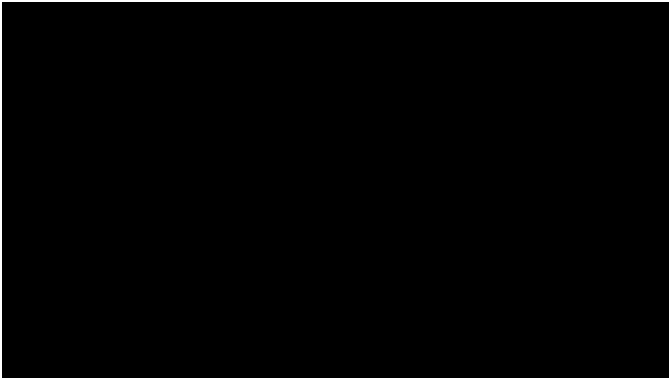
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- ### Completing a statistical sample
- Percentage of results at random
 - 20 percent of all units randomly selected to check quality
 - Must be completed on a consistent basis throughout the project
 - Statistical sampling can reduce the costs of quality control
 - Results can be mixed without an adequate testing plan and schedule

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Risk in project management

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Risk in project management

- Risk is uncertain
- Positive and negative risk effects
- Risk processes
 - Risk identification
 - Qualitative risk analysis
 - Quantitative risk analysis
 - Risk response planning
 - Risk monitoring and controlling

140

Perform qualitative risk analysis

- Prioritizing individual project risks for further analysis or action
- Assessment of probability of occurrence and impact
- Focuses efforts on high-priority risks
- Fast and quick
- Not all risks need qualitative risk analysis

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Probability-impact matrix

Risk	Probability	Impact	Risk Score
Data loss	Low	High	Moderate
Network speed	Moderate	Moderate	Moderate
Server downtime	High	Low	Moderate
Email service down	Low	Low	Low

Each identified risk

Subjective score

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Preparing for quantitative risk analysis

- Qualitative risk analysis "qualifies" the risks
- Examines and prioritizes the risks based on probability
- Impact on the project
- Broad approach to ranking risks by priority.
- Facilitator aware of perception and bias

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Responding to negative risks

- Escalating
- Avoiding
- Transferring
- Mitigating
- Accepting

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Managing positive risks

- Escalating
- Exploiting
- Sharing
- Enhancing
- Accepting

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Risk register updates

- Response strategies
- Specific actions
- Triggers, warning signs, conditions
- Budget, schedule
- Risks, owners, responsibilities
- Contingency plans
- Fallback plans

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Managing risks

- Residual risks
- Secondary risks
- Risk response contracts
- Justifying risk reduction

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Justifying risk reduction

- Additional time or monies are typically needed
- Determine if the solution is worth tradeoffs
- Probability and impact assessment

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Cost of prevention or risk of occurrence

- Cost of preventing the risk
- Cost of responding to risk
- Cost savings if risk doesn't happen rather than mitigation
- If risk does happen, the project will lose
- Cost inherent in the project delay
- Judgment of solving the risk to reduce impact
- Time and cost

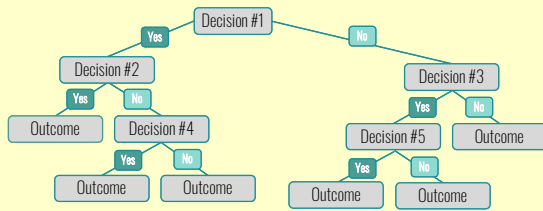
149

Using a decision tree

- Examines the cost and benefits of each decision's outcomes and weighs the probability of success for each of the decisions
- The purpose of the decision tree is to make a decision, calculate the value of that decision, or determine which decision costs the least

150

Decision Tree



151

Results of implementing risk responses

- Issue log: any issues created are recorded in the issue log
- Lessons learned register: updated with information
- Project team assignments: necessary resources allocated
- Risk register: updated to reflect any changes
- Risk report: updated to reflect changes to project risk exposure

152

Monitor risks

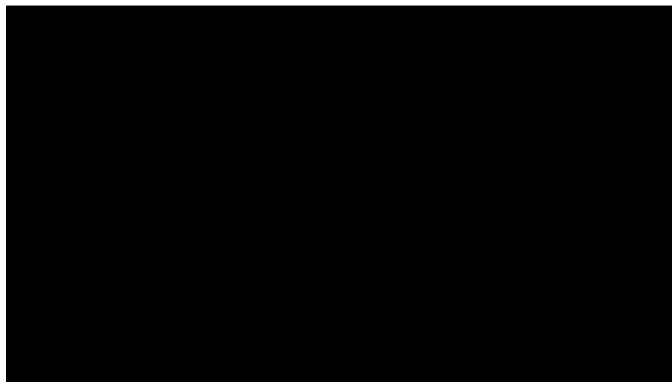
- Implementing risk response plans
- Tracking identified risks
- Monitoring residual risks
- Evaluating risk response effectiveness

153

Results of risk monitoring

- Work performance information
- Change requests
 - Corrective actions
 - Preventive actions
- Project management plan updates
- Project document updates
- Organizational process assets updates

154



155

Purpose for the plan

156

Purpose for the plan

- Communicates intent of the project
- Serves as a guide for the project manager
- Provides project structure
- Provides documentation
- Provides baselines (scope, cost, schedule, performance)

157

Developing the project plan

- Triple constraints of project management
- Iron triangle
- Inverted iron triangle
- Balance time, cost, and scope constraints

158

Planning participants

Participant	Contribution
Project manager/ScrumMaster	Leadership, facilitation, organization, direction, expert judgment
Project team members	Knowledge of project work, time estimating, schedule, risk assessment, expert judgment
Customers/Product Owner/BA	Objectives, quality requirements, influence on budget and schedule
Management/Project Sponsor	Budget, resources, project management methodology, quality requirements, project plan approval

159

Typical project management plan

- Scope management plan
- Configuration management plan
- Cost management plan
- Schedule management plan
- Quality management plan
- Resource management plan
- Communications management plan
- Risk management plan

160

Typical project management plan

- Procurement management plan
- Stakeholder engagement plan
- Scope baseline
- Cost baseline
- Schedule baseline
- Performance measurement baseline

161

Predictive project documents

- Activity attributes
- Activity cost estimates
- Activity duration estimates
- Activity list
- Activity resource requirements
- Agreements
- Basis of estimates
- Change log
- Change requests
- Forecasts (costs, schedule)

162

Predictive project documents

- Procurement documents
- Procurement statement of work
- Issue log
- Milestone list
- Project funding requirements
- Project schedule
- Project calendars
- Project charter
- Project statement of work
- Quality checklists

163

Predictive project documents

- Project schedule network diagrams
- Project staff assignments
- Requirements documentation
- Requirements traceability matrix
- Quality control measurements
- Quality metrics
- Risk register
- Schedule data
- Resource breakdown structure
- Resource calendars

164

Predictive project documents

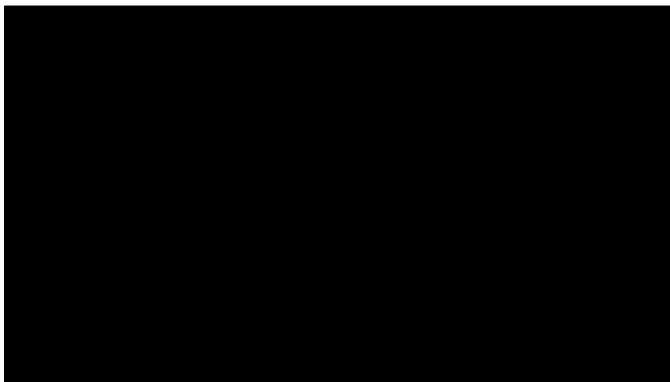
- Stakeholder register
- Team performance assessments
- Seller proposals
- Source selection criteria
- Work breakdown structure (WBS)
- Work performance reports
- Work performance data

165

Adaptive project documents

- Product vision: describes the overall goals and objectives
- Product roadmap: outlines key milestones and deliverables
- User stories: descriptions of the features and functionality that the project team is building, written from the perspective of the end user
- Release plans: features and functionality delivered in each release
- Iteration plans: work that will be done in each iteration

166



167

Distinguish between the different deliverables of a project management plan versus a product management plan

168

Project management plan

- What the project will create
- How the project will deliver
- How the project will operate

169

Product management plan

- Product lifecycle and management
- Roadmap for product team
- Alignment with organizational values

170

Product management plan components

- Product vision: overall goals and objectives of the product
- Target market: customer base, including demographic information and needs/pain points
- Product positioning: how the product will be positioned, including any unique features or benefits
- Product roadmap: outlines the key milestones and deliverables for the product over time
- Marketing and sales strategy: promote and sell the product (continued)

171

Product management plan components

- Product development plan: steps in developing the product, including research and development
- Product launch: marketing and sales efforts
- Budget and resources: resources (e.g., time, money, personnel) and a budget of projected costs
- Product metrics: key performance indicators (KPIs) to measure success
- Risks and issues: risks or issues that may impact the product and a plan for how to address them

172

Project and product differences

- A project is a temporary effort undertaken to create a unique product, service, or result
- A product is a tangible or intangible object created as a result of a project

173

Project and product differences: Duration

- A project is temporary in nature and is completed once the project's objectives have been achieved
- A product has a longer lifespan and may continue to be used or consumed for an extended period

174

Project and product differences: Scope

- A project has a specific scope that outlines the work that needs to be done
- A product has a much broader scope and may include a wide range of features and functionality

175

Project and product differences: Team

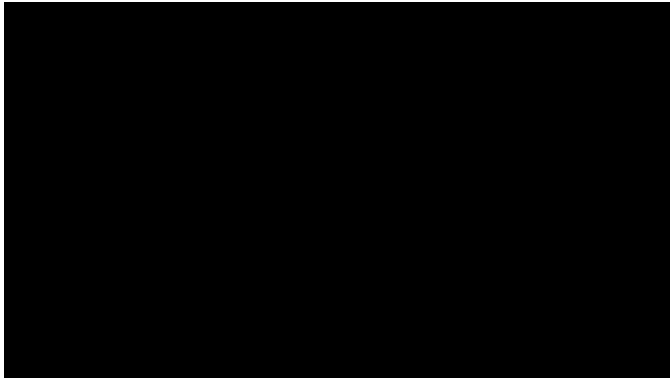
- A project team is typically made up of individuals who are brought together specifically for the purpose of completing the project
- A product team may be a permanent group of individuals responsible for managing the product throughout its lifecycle

176

Project and product differences: Deliverables

- A project delivers specific outputs, such as a product or service, at the end of the project
- A product is the primary deliverable and is intended for use or consumption by the end user

177



178

Distinguish differences between a milestone and a task duration

179

Distinguish differences between a milestone and a task duration

- Milestones are timeless markers that signal progress
 - Foundation is poured
 - Presentation slide deck created
 - Server install completed

180

Distinguish differences between a milestone and a task duration

- Tasks are the work required to reach the milestone
 - Excavate the job site for the concrete
 - Create an outline for the slide deck
 - Configure the server hardware

181

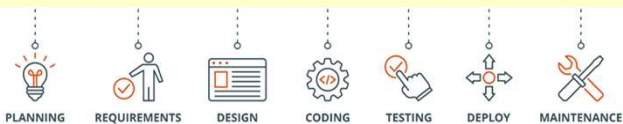
Distinguish differences between a milestone and a task duration

- Literally refers to a stone marker to indicate the distance traveled
- Project milestones are used to mark the progress made on a project
- Term "milestone" in project management dates to the early 20th century
- Originally used to describe a key event or achievement
- Marked a major turning point or significant progress

182

Project life cycle

- Unique to each project
- Duration of the project
- Phases of construction versus phases of IT projects



183

Project phases

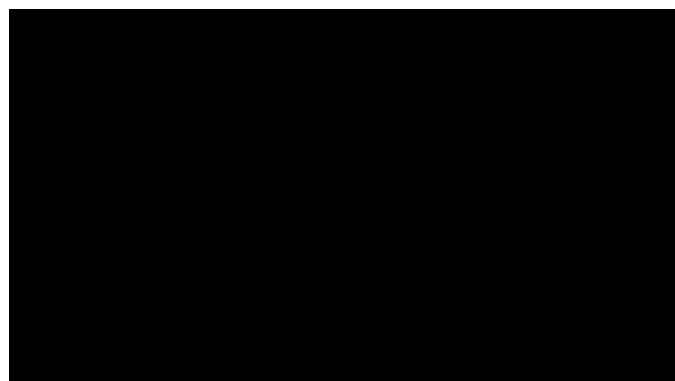
- Phases result in key deliverables
- Phase names describe work:
 - Foundation
 - Framing
 - Interior
 - Exterior
 - Milestones often linked to phases

184

Task list

- Project scope statement
- Work breakdown structure
- Work packages
- Activity list
- 8/80 rule

185



186

Determine the number and type of resources in a project

187

Determine the number and type of resources in a project

- Human resources
 - People do the work
 - Sometimes called labor
- Physical resources
 - Equipment, hardware, software
 - Facilities

188

Scope of the project

- Key factor when determining resources needed
- Larger, more complex projects will typically require more resources than a smaller, simpler project
- Risks that may impact the number of resources needed

189

Timing and duration of the project

- Schedule and duration can impact the number of resources needed
- Tight deadlines may require additional resources
- Longer projects may require fewer resources, as the work can be spread out over a longer time

190

Available resources and constraints

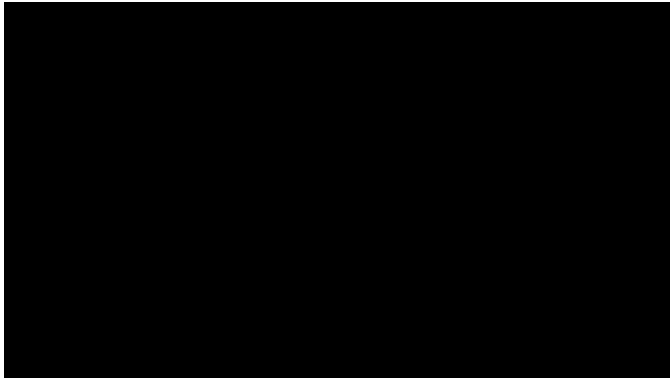
- Availability of resources and constraints
- Other projects
- Operations
- Specific skillset

191

Project budget

- Budget can affect the resources
- Limited budget requires planning how resources are allocated
- Identify resources that are most essential to the project

192



193

Use a stakeholder register in a given situation

194

Introduction to the stakeholder register

- Definition of stakeholder register
- Purpose of stakeholder register in project management
- How a stakeholder register helps identify and manage stakeholders
- Steps involved in creating and maintaining a stakeholder register

195

Identifying stakeholders

- Identify all individuals and groups who have an interest in or are affected by the project
- Determine their level of influence and interest in the project
- Assign a stakeholder classification (e.g. high, medium, low) based on influence and interest
- Document all stakeholders in the stakeholder register

196

Analyzing stakeholder requirements

- Determine the needs and expectations of each stakeholder
- Analyze the impact of the project on each stakeholder
- Prioritize stakeholders based on their level of influence and impact
- Document all stakeholders' requirements and impact in the stakeholder register

197

Developing stakeholder engagement plan

- Identify the most effective communication methods for each stakeholder
- Develop a plan for engaging and communicating with stakeholders throughout the project
- Identify key stakeholders and assign a primary point of contact
- Document the stakeholder engagement plan in the stakeholder register

198

Tracking stakeholder engagement

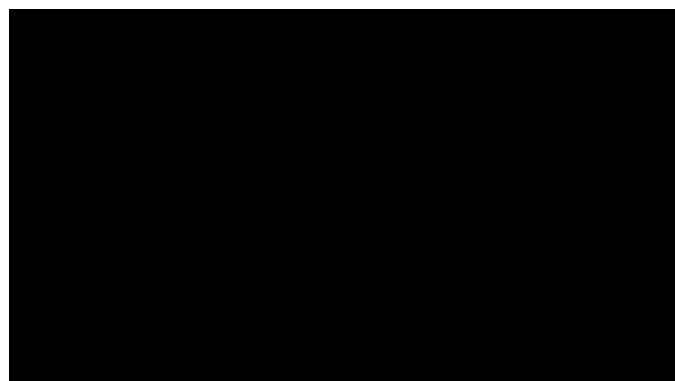
- Monitor and document stakeholder engagement throughout the project
- Update the stakeholder register with any changes in stakeholder involvement or requirements
- Use the stakeholder register to identify potential risks and issues
- Implement strategies for addressing any identified risks or issues

199

Maintaining the stakeholder register

- Review and update the stakeholder register regularly
- Ensure that the stakeholder register is kept up-to-date
- Reference tool for managing engagement throughout the project

200



201

Explain project closure and transitions

202

Introduction to project closure and transitions

- Definition of project closure and transitions
- Importance of properly closing and transitioning a project
- Common challenges and pitfalls in project closure and transitions
- Steps involved in a successful project closure and transition process

203

Reviewing and evaluating project outcomes

- Review the project's performance against its goals and objectives
- Evaluate the project's successes and challenges
- Document lessons learned and best practices for future projects
- Prepare a final project report detailing the project's outcomes

204

Closing out project deliverables and resources

- Complete and deliver all final project deliverables
- Release any resources that were dedicated to the project
- Document and archive project materials
- Disband the project team and release team members to their respective organizations

205

Transferring ownership of deliverables and resources

- Identify the parties responsible for maintaining and using the project's deliverables and resources
- Coordinate the transfer of ownership and responsibility for deliverables and resources
- Provide training and support as needed to ensure a smooth transition
- Document the transfer of ownership and responsibility in the project record

206

Finalizing financial and legal matters

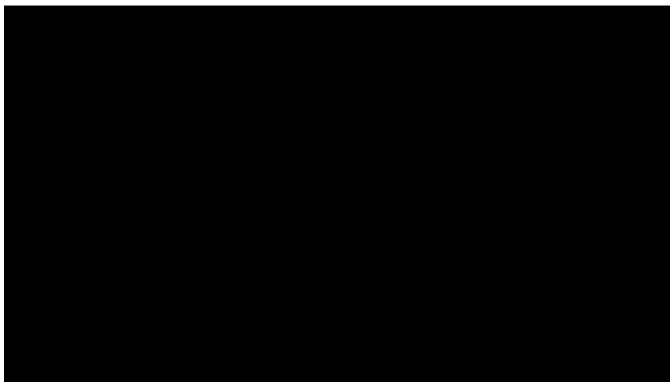
- Review and finalize all financial accounts related to the project
- Settle any outstanding debts or liabilities
- Review and close out any contracts or agreements related to the project
- Document all financial and legal matters in the project record

207

Celebrating project successes and recognizing contributions

- Host a celebration or recognition event for the project team
- Acknowledge and thank key stakeholders for their contributions
- Share the project's outcomes and successes with the broader organization
- Document the celebration and recognition efforts in the project record

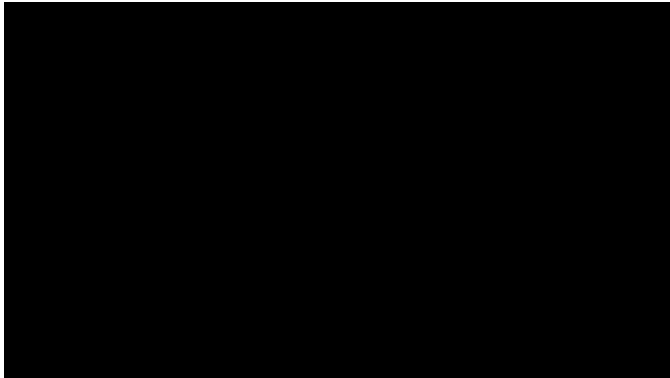
208



209

CAPM Coach

210



211

Section conclusion

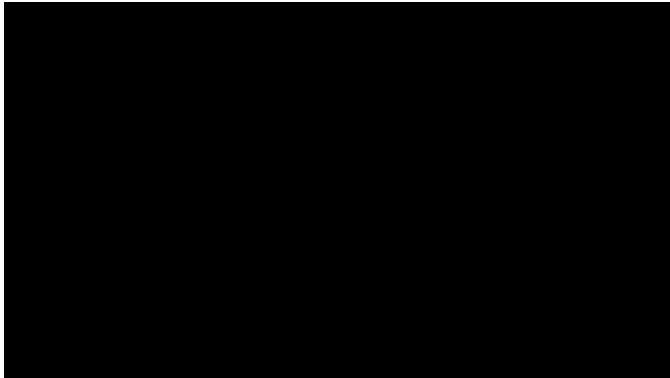
Demonstrate an understanding of project management planning

212

Section overview

Demonstrate an understanding of project roles and responsibilities

213



214

Project management roles

215

Project manager

- Responsible for overseeing the entire project
- Setting goals
- Developing a plan
- Ensuring the project stays:
 - On schedule
 - Within budget
 - Monitoring KPIs

216

Project sponsor

- A key stakeholder in a project
- Ensures the success of the project they are responsible for providing:
 - Resources
 - Support
 - Vision
 - Definition of business value

217

Product Owner

- Representing the interests of the stakeholders
- Determining the features and priorities for the product
- Work closely with the development team to ensure:
 - The product is delivered on time
 - Meets the desired business goals

218

Project team

- Developing and implementing a project plan
- Monitoring progress
- Managing budgets
- Stays on track
- Meets goals

219

Scrum Master

- Facilitating the scrum process
- Ensuring the team follows the agile framework
- Help the team:
 - Stay focused
 - Deliver high-quality products

220

Team lead

- Design
- Development
- Work closely with the PM to ensure their part of the project is:
 - Completed on time
 - To required specifications
- Examples of team leads include:
 - Tech leads
 - Design leads

221

Developer(s)

- Building
- Implementing
- Writing code
- Designing interfaces
- Configuring systems

222

Tester(s)

- Responsible for verifying the project:
 - Meets the specified requirements
 - Is of high quality
- May involve:
 - Writing and executing test cases
 - Reporting any issues

223

Designer(s)

- Visual designs
- User experience designs
- May include creating:
 - Wireframes
 - Prototypes
 - High-fidelity mockups

224

Writer(s)

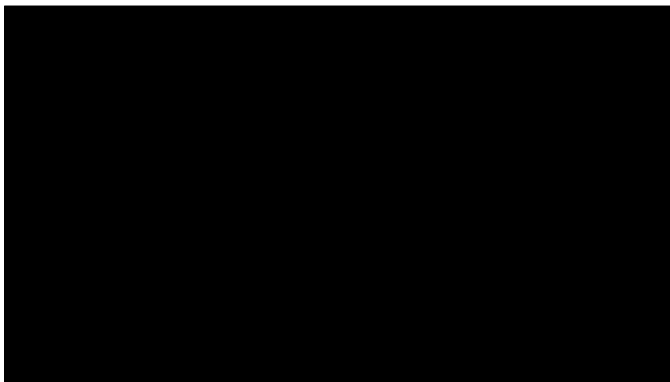
- Responsible for creating written content:
 - User manuals
 - Help documentation
 - Marketing materials

225

Stakeholders

- Individuals or groups that have a vested interest in the project
- Project sponsor
- Customers
- Users
- May involve:
 - Providing input and feedback on the project
 - Approving deliverables

226



227

Project manager role

228

Project manager role

- Planner
- Team leader
- Resource manager
- Progress monitor
- Communicator
- Scope controller
- Closer

229

Project manager responsibilities

- Developing a project plan
- Leading the project team
- Managing project resources
- Monitoring progress
- Communicating with stakeholders
- Controlling the project scope
- Closing the project

230

Developing a project plan

- Defining project scope
- Setting goals and objectives
- Determining tasks and milestones
- Creating a budget

231

Leading the project team

- Communicating with team members
- Assigning tasks and responsibilities
- Providing support and guidance

232

Managing project resources

- Budget
- Equipment
- Personnel

233

Monitoring progress

- Team members
- Stakeholders
- Identifying potential:
 - Issues
 - Roadblocks

234

Communicating with stakeholders

- Progress
- Changes
- Issues

235

Controlling the project scope

- Scope creep is avoided
- Any changes to the scope are carefully managed

236

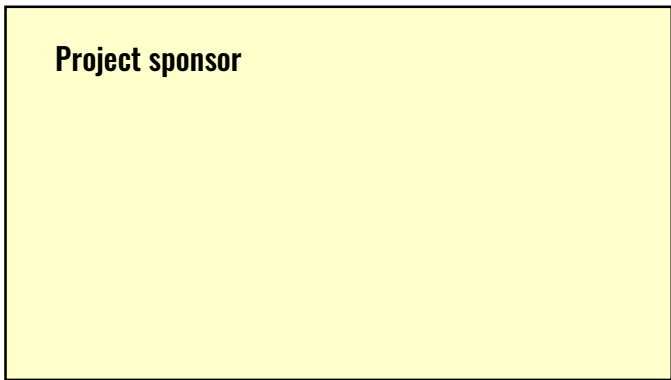
Closing the project

- Collecting and analyzing data
- Writing a report
- Completing a final review

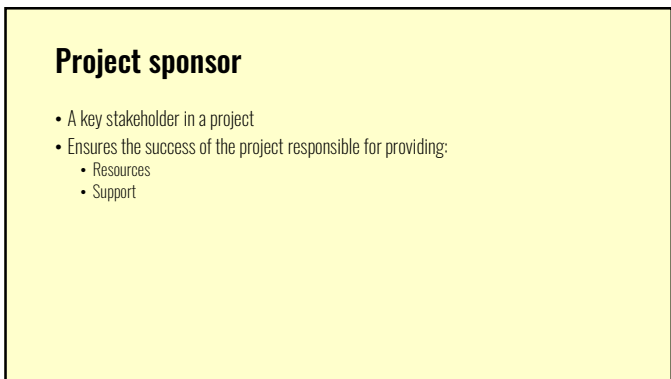
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238



239



240

Project sponsor roles

- Champion
- Decision-maker
- Communicator
- Leader
- Business representative
- Resource provider

241

Champion

- Serves as a champion for the project
- Defines business value
- Establishes project importance

242

Decision-maker

- Has authority to make key decisions about the project
- Approves scope modifications
- Defines and can approve budget changes

243

Communicator

- Progress
- Changes
- Issues

244

Leader

- Leadership
- Direction

245

Business representative

- Ensures the project is aligned with business goals
- Project must meet organizational objectives

246

Resource provider

- Budget
- Personnel

247

Project sponsor responsibilities

- Providing financial and strategic support
- Acting as a decision-maker
- Communicating with stakeholders
- Facilitating communication and collaboration
- Providing leadership and direction
- Ensuring the project aligns with business goals

248

Communicating with stakeholders

- Progress
- Changes
- Issues

249

Facilitating communication and collaboration

- Ensures everyone is working towards the same goals
- Encourages open communication and collaboration:
 - Team members
 - Stakeholders

250

Providing leadership and direction

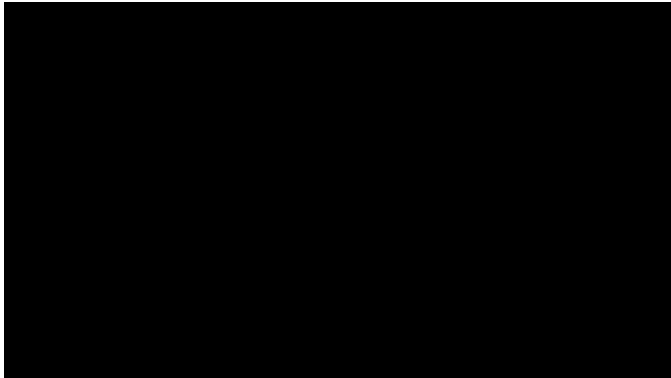
- Guidance
- Direction

251

Ensuring the project aligns with business goals

- Business goals
- Objectives

252



253

Comparison of project manager and project sponsor roles

254

Comparison of project manager and project sponsor roles

Project manager

- Planner
- Team leader
- Resource manager
- Progress monitor
- Communicator
- Scope controller
- Closer

Project sponsor

- Champion
- Decision-maker
- Communicator
- Leader
- Business representative
- Resource provider

255

Project manager responsibilities

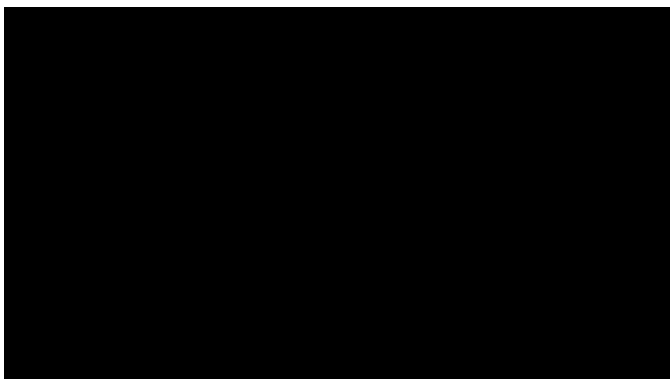
- Developing a project plan
- Leading the project team
- Managing project resources
- Monitoring progress
- Communicating with stakeholders
- Controlling the project scope
- Closing the project

256

Project sponsor responsibilities

- Providing financial and strategic support
- Acting as a decision-maker
- Communicating with stakeholders
- Facilitating communication and collaboration
- Providing leadership and direction
- Ensuring the project aligns with business goals

257



258

Project team roles

259

Project team roles

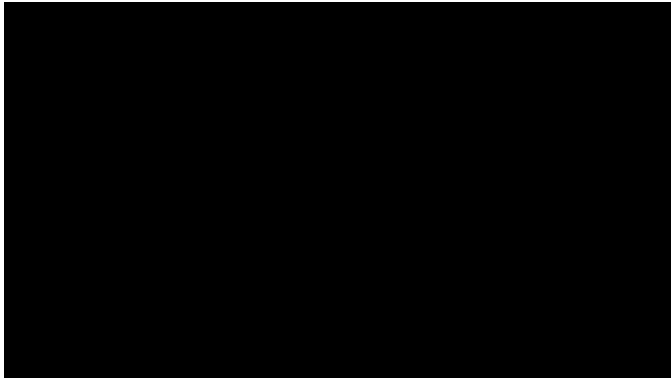
- Project manager
- Team leader
- Team members
- Stakeholders
- Sponsors

260

Project team responsibilities

- Developing and implementing a project plan
- Managing the project budget
- Monitoring progress
- Communicating with stakeholders
- Managing risks
- Ensuring quality control
- Managing team performance
- Reviewing and adjusting the project plan as needed

261



262

Project team responsibilities

263

Project team roles

- Project manager
- Team leader
- Team members
- Stakeholders
- Sponsors

264

Project sponsor roles

- Providing financial and strategic support
- Acting as a decision-maker
- Communicating with stakeholders
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265

Project manager responsibilities

- Developing and implementing a project plan
- Managing the project budget
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- Managing team performance
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266

Project sponsor responsibilities

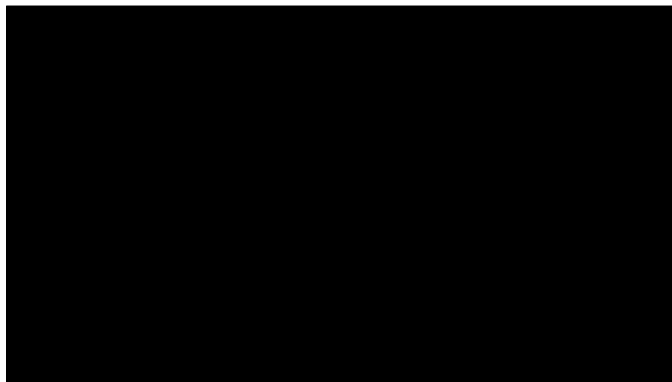
- Providing financial and strategic support
- Acting as a decision-maker
- Communicating with stakeholders
- Facilitating communication and collaboration
- Providing leadership and direction
- Ensuring the project aligns with business goals

267

Project team responsibilities

- Executing the project plan
- Completing assignments
- Collaboration
- Reporting progress
- Controlling quality

268



269

Importance of the project manager role

270

Importance of the project manager role

- Critical role in the success of a project
- Responsible for overseeing the entire project
- Keeping project on track
- Monitoring project goals

271

Project manager role

- Initiator
- Negotiator
- Listener
- Coach
- Working member
- Facilitator

272

Initiator

- Responsible for initiating the project plan, which involves:
 - Development
 - Implementation

273

Negotiator

- Stakeholders
- Team members
- Vendors
- Other parties

274

Listener

- Feedback
- Ideas
- Information should be readily received from:
 - Team members
 - Stakeholders

275

Coach

- Support
- Guidance
- Help team develop their skills
- Promote professional growth

276

Working member

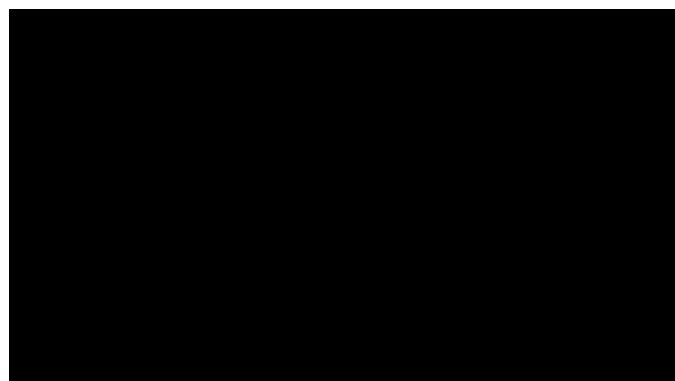
- Be willing to roll up their sleeves and contribute to the project
- Not just delegate tasks to others

277

Facilitator

- Communication
- Collaboration
- Ensure that everyone is working toward the same goals

278



279

Differences between leadership and management

280

Differences between leadership and management

- Both are important for the success of a project
- Many PMs are responsible for both tasks
- Differing roles, responsibilities, skills

281

Leadership

- The ability to inspire and guide a team to achieve a common goal
- Setting a clear vision for the project
- Motivating team members
- Decisions that align with the project's objectives
- Influence others
- Build trust
- Create a positive work culture

282

Management

- Organizing and coordinating work activities
- Allocating resources
- Tracking progress
- Plan and schedule project work
- Control project activities
- Ensure that the project is completed on time and within budget

283

Emotional intelligence (EQ) and its impact on project management

- Recognize and understand own emotions and emotions of others
- Manage own behavior and relationships effectively
- Leadership
- Management

284

Emotional intelligence (EQ) and its impact on project management

- Improved communication
- Conflict resolution
- Motivation
- Adaptability

285

Conflict resolution

- Emotions
- Perspectives

286

Motivation

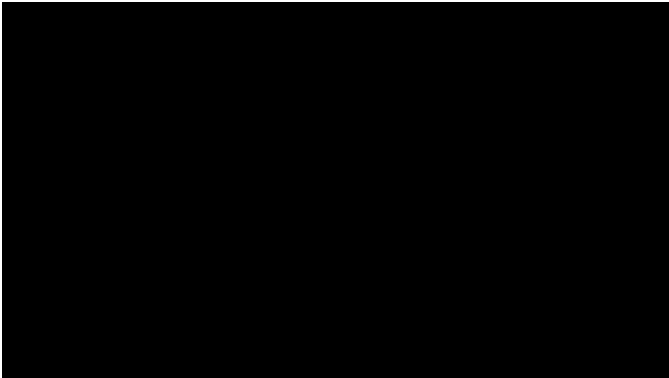
- Motivate
- Inspire
- Create a positive work culture

287

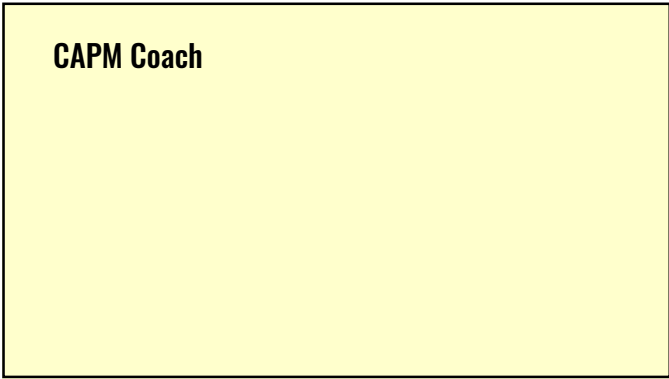
Adaptability

- Adapt to changing circumstances
- Handle stress more effectively

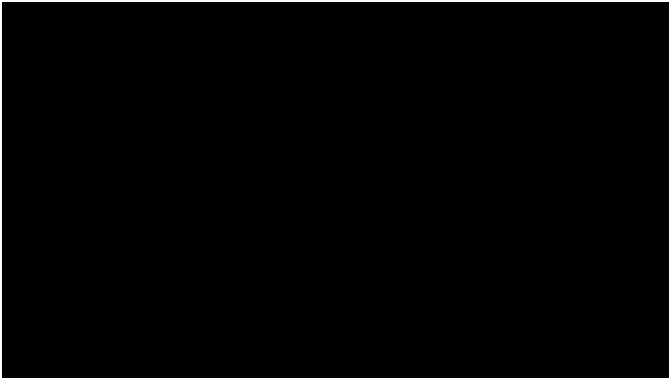
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289



290



291

Section conclusion

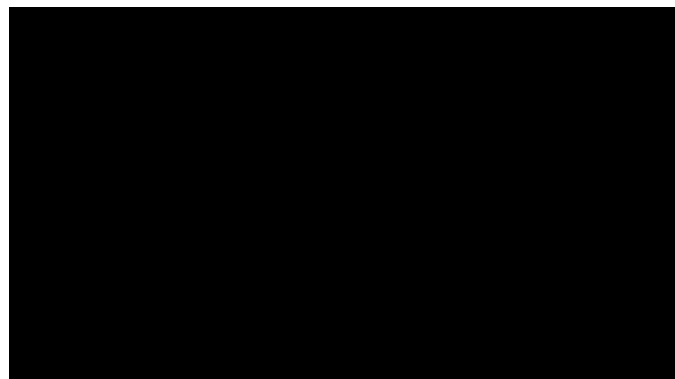
Demonstrate an understanding of project roles and responsibilities

292

Section overview

Determine how to follow and execute planned strategies or frameworks

293



294

Follow and execute planned strategies or frameworks

295

Steps to effectively follow and execute planned strategies or frameworks

- Clearly define the project scope and objectives
- Develop a project plan
- Communicate the project plan to relevant stakeholders
- Identify any potential risks or challenges
- Establish clear benchmarks and milestones
- Monitor and assess progress
- Celebrate successes and learn from setbacks

296

Clearly define the project scope and objectives

- Goals and objectives of the project
- Specific deliverables and outcomes that are expected

297

Develop a project plan

- Create a detailed plan of what needs to be completed in order to meet project objectives
- Steps and tasks
- Address all applicable knowledge areas

298

Identify any potential risks or challenges

- Anticipate any potential risks or challenges that could arise during the project
- Develop contingency plans to address them

299

Establish clear benchmarks and milestones

- Help you track your progress
- Ensure that you are on track to meet your project objectives

300

Monitor and assess progress

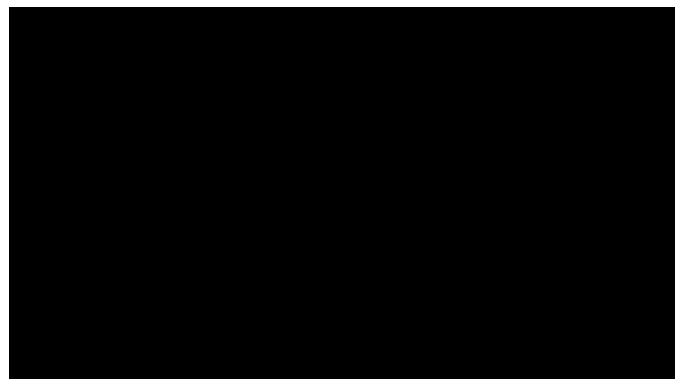
- Regularly assess your progress
- Adjust as needed

301

Celebrate successes and learn from setbacks

- Take the time to celebrate successes
- Reflect on any setbacks that may have been encountered
- Future planning
- Decision-making

302



303

Examples of how it is appropriate to respond to a planned strategy or framework

304

Examples of how it is appropriate to respond to a planned strategy or framework

- Communication
- Risk
- Performance
- Change management
- Stakeholder engagement

305

Communication

- Communication processes
- Channels
- Update relevant stakeholders on the new communication protocols
- Ensure all necessary training and support is provided

306

Risk

- Risk management plan
- Risk response plan
- Risk log

307

Performance

- Setting clear performance targets
- Monitoring progress towards meeting those targets

308

Change management

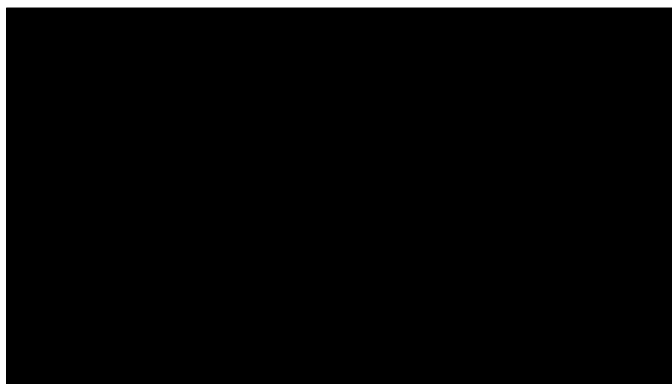
- Support
- Resources
- Doing so may help employees adapt to the new changes

309

Stakeholder engagement

- Regularly soliciting input and feedback from stakeholders
- Incorporating input into the implementation of the strategy or framework

310



311

Project initiation and benefit planning

312

Project initiation and benefit planning

- Project initiation is the process of defining and setting up a new project
- Benefit planning is the process of identifying, quantifying, and prioritizing the potential benefits that the project is expected to deliver

313

Project initiation

- Identify the need for the project
- Define the project scope and objectives
- Assemble the project team
- Develop a project plan
- Secure funding and resources

314

Identify the need for the project

- Customer demand
- Regulatory requirements
- Company goals
- Business value

315

Define the project scope and objectives

- Specific deliverables
- Outcomes that are expected

316

Assemble the project team

- Identify
- Assemble
- Selection, pre-assignment, negotiation

317

Secure funding resources

- Physical resources: materials, facilities, travel
- Budget
- Personnel
- Equipment

318

Benefit planning

- Involves identifying and quantifying the potential benefits that the project is expected to deliver
- Prioritizing those benefits based on their importance to the organization
- Project is aligned with the organization's strategic goals
- Resources invested in the project are likely to yield a positive return on investment

319

Benefit planning

- Identify the potential benefits of the project
- Quantify the potential benefits
- Prioritize the potential benefits
- Develop a benefits management plan
- Monitor and assess benefits
- Communicate the benefits

320

Benefit planning

- Organization
- Customers
- Other stakeholders
- Potential benefits could include things like:
 - Cost savings
 - Increased efficiency
 - Improved customer satisfaction
 - New revenue streams

321

Quantify the potential benefits

- Determine the potential financial value of each identified benefit
- Conducting financial analysis
- Using statistical models

322

Prioritize the potential benefits

- Expected financial return
- Likelihood of achieving the benefit
- Potential impact on the organization's strategic goals

323

Develop a benefits management plan

- Actions
- Resources

324

Monitor and assess benefits

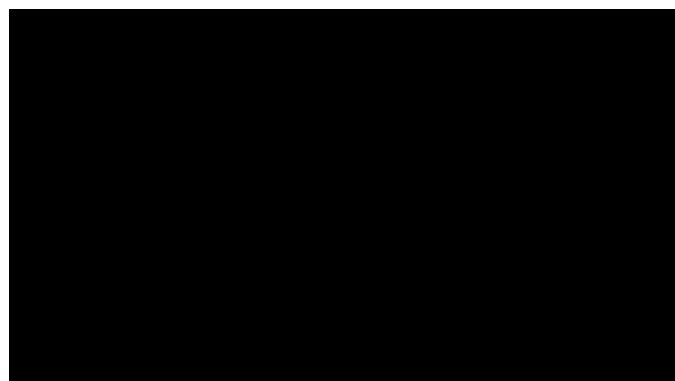
- Monitor and assess the benefits to ensure that they are being realized
- Make any necessary adjustments to the benefit management plan

325

Communicate the benefits

- Project team
- Management
- Customers
- Value of the project
- Impact on the organization

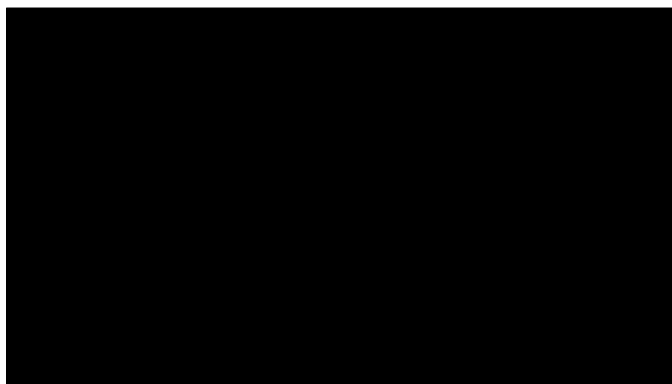
326



327

CAPM Coach

328



329

Section conclusion

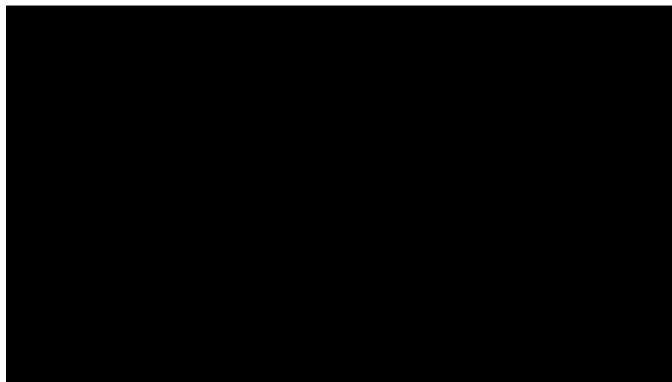
Determine how to follow and execute planned strategies or frameworks

330

Section overview

Demonstrate an understanding of common problem-solving tools and techniques

331



332

Problem-solving tools and techniques

333

Common problem-solving tools and techniques

- Brainstorming
- Root cause analysis
- Fishbone diagram
- Swot analysis
- Pareto analysis
- Decision tree
- Six sigma

334

Common problem-solving tools and techniques

- Issue tracking
- Risk assessment
- Change management
- Project reviews

335

Brainstorming

- Team members freely share ideas without criticism

336

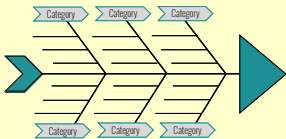
Root cause analysis

- Identify underlying causes of a problem by breaking it down into smaller components and examining each component in detail

337

Fishbone diagram

- Categories and relationships between them
- Cause and effect
- Causal factors and contributing causes



338

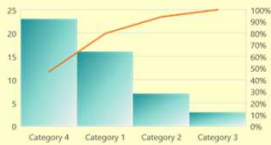
SWOT analysis

- Strengths
- Weaknesses
- Opportunities
- Threats

339

Pareto analysis

- A tool used to identify the most significant factors contributing to a problem by ranking them in order of importance



340

Decision tree

- Potential consequences of different decisions or actions, used to help make informed choices

341

Six Sigma

- A methodology that uses data and statistical analysis to identify and eliminate defects in a process with the goal of reducing variability and achieving near-perfect performance

342

Issue tracking

- Identifying and documenting problems or challenges

343

Risk assessment

- Evaluating risks that could impact a project, and developing strategies to mitigate those risks

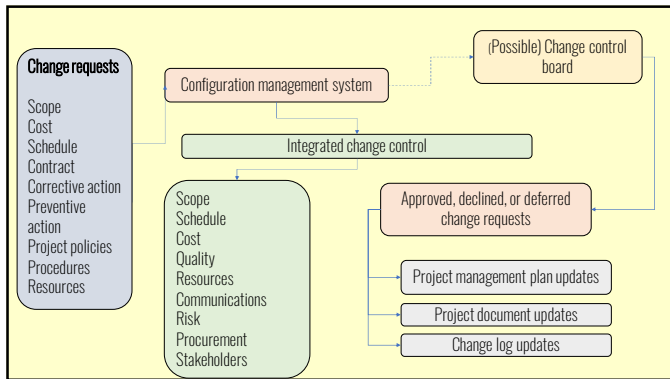
Likelihood	Consequences				
	Insignificant	Minor	Moderate	Major	Catastrophic
Certain >90% Chance	High	High	Extreme	Extreme	Extreme
Likely 50%-90% Chance	Moderate	High	High	Extreme	Extreme
Moderate 10%-50% Chance	Low	Moderate	High	Extreme	Extreme
Unlikely 3%-10% Chance	Low	Low	Moderate	High	Extreme
Rare <3% Chance	Low	Low	Moderate	High	High

344

Change management

- Managing changes to the project scope, schedule, or budget, and ensuring that any necessary approvals and documentation are obtained

345

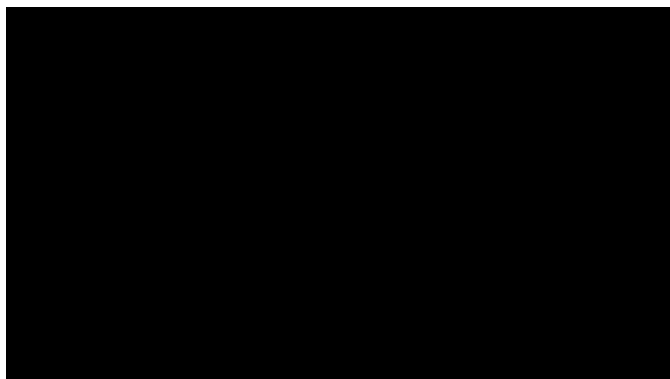


346

Project reviews

- Regularly reviewing the progress of a project
- Identifying any problems or issues
- Taking corrective action

347



348

Evaluate the effectiveness of a meeting

349

Evaluate the effectiveness of a meeting

- It achieved its goals
- Was participatory and efficient
- Had clear follow-up and action items
- Received positive feedback from attendees

350

Consider participation and engagement

- Everyone has an opportunity to contribute
- One-way presentation with little interaction

351

Assess the meeting efficiency

- Well-organized
- Focused
- Did it feel:
 - Disjointed
 - Unproductive

352

Consider follow-up and action items

- Clear next steps
- Identified action items
- Follow-ups

353

Ask for feedback

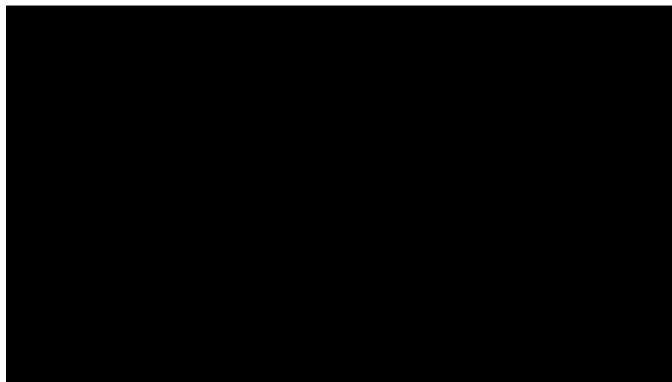
- Comments
- Suggestions

354

Reflect on the meeting

- What worked well?
- What could be improved?
- What was the result?

355



356

Purpose of common types of meetings

357

Purpose of common types of meetings

- Focus groups
- Standup meetings
- Brainstorming
- Retrospectives
- Design thinking workshops

358

Focus groups

- Small, facilitated discussions with a group of people
- Typically involve 8-12 participants
- Gather qualitative data
- Insights
- Perspectives
- Ideas

359

Standup meetings

- Short, daily meetings that are typically held in a standing position
- Daily Scrum
- Daily standups
- Purpose is to provide a quick update on:
 - Progress of a project or task
 - Identify any obstacles that need to be addressed
 - Coordinate work for the day

360

Brainstorming

- Encourage creative thinking
- Generate a diverse range of potential solutions

361

Retrospectives

- Project
- Iteration
- What went well?
- What could be improved?

362

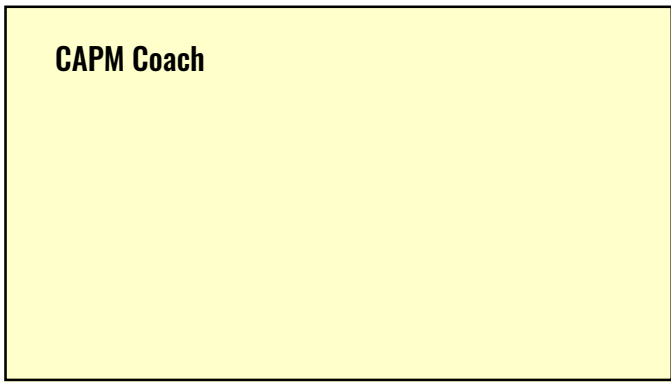
Design thinking workshops

- Focused on problem-solving and innovation
- Identify user needs
- Generate ideas
- Prototype and test potential solutions
- Continuous improvement
- Innovation

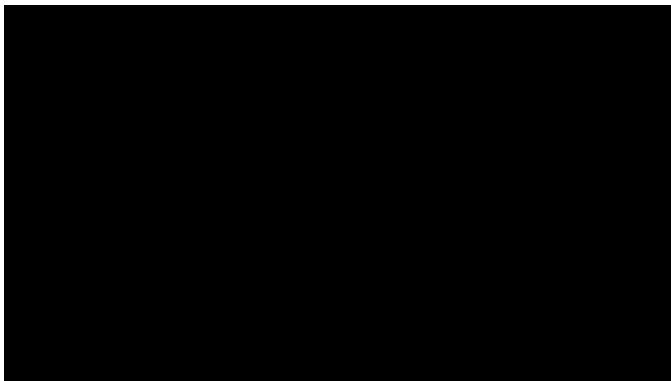
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364



365



366

Section conclusion

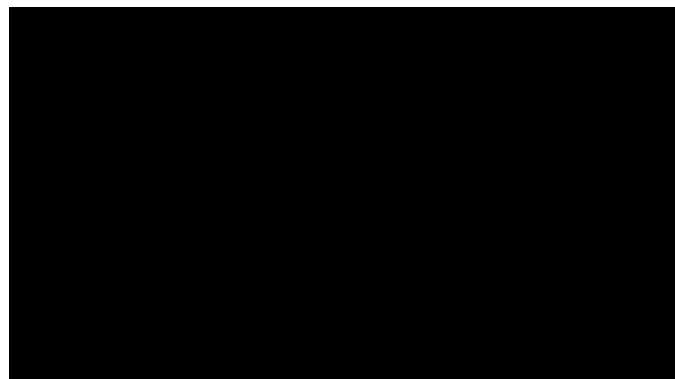
Demonstrate an understanding of common problem-solving tools and techniques

367

Section overview

Explain when it is appropriate to use a predictive, plan-based approach

368



369

Predictive, plan-based approach

370

Predictive, plan-based approach

- Trying to achieve a specific goal
- There is a clear understanding of the steps needed to reach that goal
- Involves:
 - Creating a plan outlining the steps that will be taken to achieve the goal
 - Using data and predictions about the future to guide the execution of the plan

371

Appropriate use of a predictive, plan-based approach

- When a goal is well-defined, and there is a clear path to achieving it
- When the future is somewhat predictable
- When you have limited resources

372

Well-defined goal

- Efficiently and effectively
- Clear understanding of what you want to accomplish
- Plan to achieve it

373

Unpredictable future

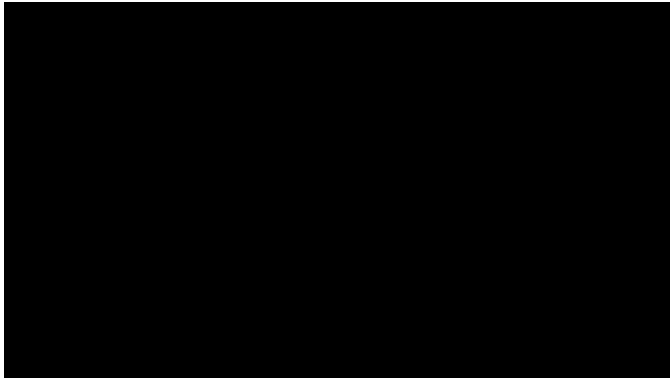
- Challenges
- Opportunities
- A good understanding of the factors that will influence the outcome of efforts
- A reasonable way to make predictions about how factors will change over time

374

Limited resources

- Prioritizing the most important tasks
- Allocating resources
- Time
- Money
- Equipment
- Team members

375



376

**Suitability of a predictive, plan-based approach
for the organizational structure**

377

**Suitability of a predictive, plan-based approach
for the organizational structure**

- Goal-oriented
- Clear hierarchy of responsibilities
- Strong leadership and management
- Efficient communication and collaboration

378

Suitability of a predictive, plan-based approach for the organizational structure

- Tasks and activities are aligned with the overall goals of the organization
- Resources are used efficiently to achieve those goals

379

Characteristics

- Clearly defined roles and responsibilities
- Strong leadership and management
- Efficient communication and collaboration

380

Clearly defined roles and responsibilities

- Tasks
- Activities
- Identify who is responsible for each task
- Create a plan that outlines how tasks will be completed

381

Strong leadership and management

- Plan executed effectively
- Deviations are addressed
- Clear chain of command
- Monitoring progress and adjusting

382

Effective communication and collaboration

- Coordinate efforts
- Ensure that tasks are completed on time and as expected
- Well-suited to using a predictive, plan-based approach

383

Predictive: Suitability for the organizational structure

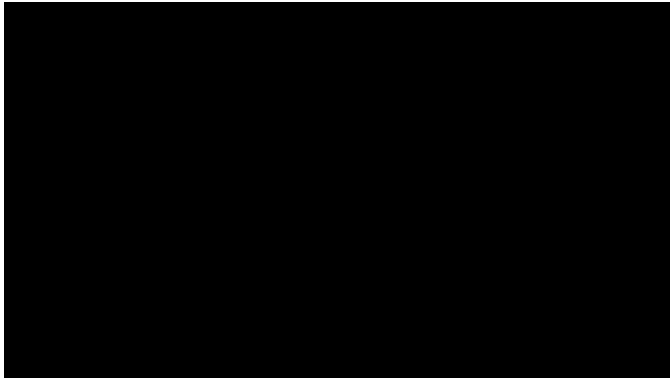
Suitable

- Matrix structure
- Hierarchical
- Clear chain of command
- Centralized decision-making authority
- Clear communication
- Coordination of plans and actions

Unsuitable

- Virtual
- Collocated
- Decentralized
- Emphasis on:
 - Teamwork
 - Collaboration
 - Adaptability to changing circumstances

384



385

**Defining goals and objectives in a predictive,
plan-based project - example**

386

**Defining goals and objectives in a predictive,
plan-based project - example**

- Goal: launch a new product
- Objectives:
 - Develop a product that meets customer needs and expectations
 - Achieve profitability within the first year of launch
 - Secure at least 50 percent market share within the first two years of launch
 - Achieve a positive return on investment within the first three years of launch

387

Defining goals and objectives in a predictive, plan-based project - example

- Direction
- Focus
- Decision-making
- Resource allocation

388

Developing plans and schedules in a predictive, plan-based project - example

- Developed and launched
- The management team may develop the following plans and schedules:
 - Product development plan
 - Marketing plan
 - Production schedule
 - Budget

389

Developing plans and schedules in a predictive, plan-based project - example

- All necessary tasks are completed in a timely and efficient manner
- Resources are allocated appropriately to support the project

390

Monitoring progress in a predictive, plan-based project - example

- Regular check-ins
- Progress reviews

391

Monitoring progress in a predictive, plan-based project - example

- Reviewing the project schedule
- Reviewing the budget
- Gathering feedback from team members
- Reviewing market trends and customer feedback

392

Monitoring progress in a predictive, plan-based project - example

- Identify any issues or challenges that may arise
- Take corrective action as needed to keep the project on track

393

Adjusting plans as needed in a predictive, plan-based project - example

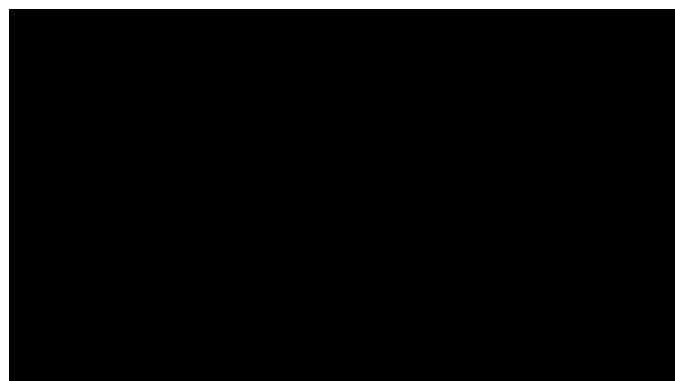
- Delays in procuring raw materials
- Issues with equipment
- Adjust the budget if:
 - Unexpected costs arise
 - Changes in the marketplace

394

Adjusting plans as needed in a predictive, plan-based project - example

- Being flexible
- Adapting to changes as needed

395



396

Goals vs objectives

397

Goals vs objectives

Goals

- Broad statements that describe what an organization or project is working towards

Objectives

- Specific, measurable targets that help to achieve the overall goals

398

Tasks vs deliverables

Tasks

- Individual steps or activities that are required to complete a project

Deliverables

- Specific outputs or results that are produced as a result of completing those tasks

399

Plans vs schedules

Plans

- Outline the steps and tasks that are required to achieve the goals and objectives of a project

Schedules

- Provide a timeline for completing those tasks and show when each task is expected to be completed

400

Resources vs budget

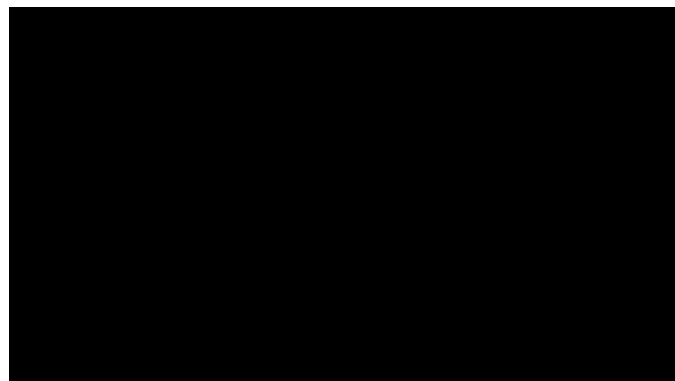
Resources

- Personnel, materials, and equipment that are needed to complete a project

Budget

- Financial plan for a project, including the costs of the resources needed and any other expenses

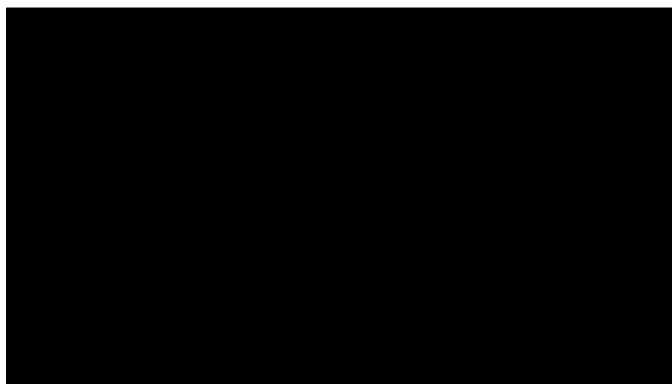
401



402

CAPM Coach

403



404

Section conclusion

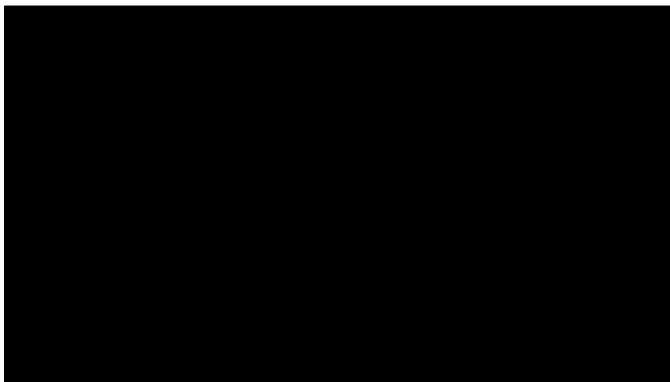
Explain when it is appropriate to use a predictive, plan-based approach

405

Section overview

Demonstrate an understanding of a project management plan schedule

406



407

Project management plan schedule

408

Project management plan schedule

- Task and milestone start dates and end dates
- Dependencies between tasks
- Allocate resources effectively
- Ensure the project stays on track

409

Schedule presentation: Gantt chart

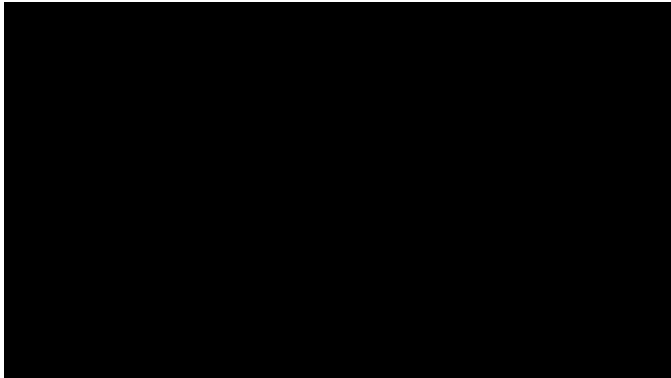
- Tasks
- Milestones
- Which tasks are coming up
- How long tasks will take
- Which resources are required to complete tasks

410

Project management plan schedule

- Keep the project on track
- Identify potential issues
- Make necessary adjustments to the plan

411



412

Critical Path Method (CPM)

413

Critical Path Method (CPM)

- The longest path through the project
- The least amount of float or flexibility
- Resources
- Efforts
- Important tasks

414

Critical Path Method (CPM) steps

- Define the project tasks
- Estimate the duration of each task
- Identify the dependencies between task
- Draw the project network diagram
- Determine the critical path
- Monitor progress and adjust the plan as needed

415

Define the project tasks

- Identify all the tasks that need to be completed in order to complete the project

416

Estimate the duration of each task

- Determine how long each task will take to complete

417

Identify the dependencies between tasks

- Determine which tasks must be completed before other tasks can begin

418

Draw the project network diagram

- Tasks
- Dependencies

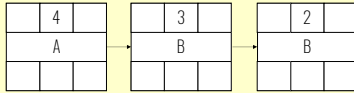
419

Determine the critical path

- Examining the project network diagram
- Calculating the length of the longest path through the project

420

Find the duration of each path



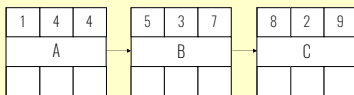
ABC = 9 days

421

Find the early start and early finish

All projects start on Day 1

$(ES + du) - 1 = EF$



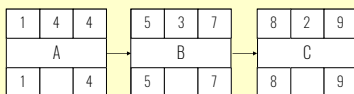
ABC = 9 days

422

Find the late start and late finish

All projects end on the latest day possible.

$(LF - du) + 1 = LS$



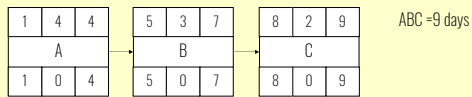
ABC = 9 days

423

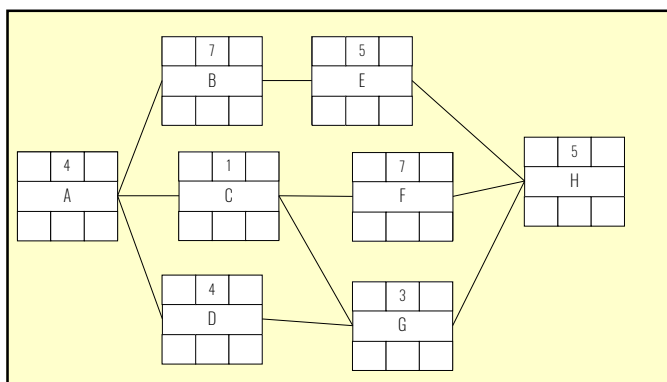
Expose the float

Critical path activities have no float

EF-LF or ES-LS = float



424

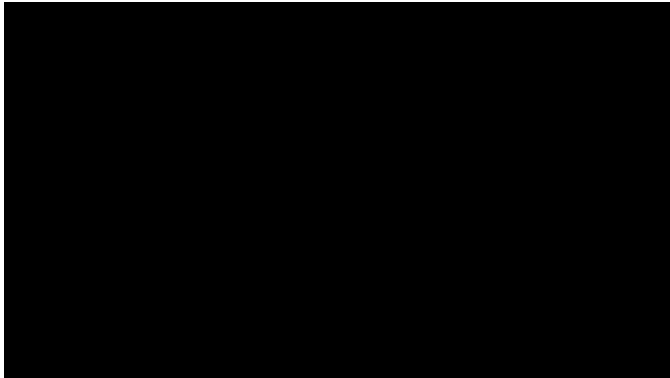


425

Monitor progress and adjust the plan as needed

- Monitor the status of the critical path tasks
- Make any necessary adjustments to the plan
- Ensure that the project stays on track

426



427

Schedule variance (SV)

428

Schedule variance (SV)

- A measure of how far a project's schedule has deviated from the original plan
- Calculated by taking the difference between:
 - The planned duration of a task or phase
 - The actual duration of the task or phase
 - Expressing it as a percentage of the planned duration

429

Schedule variance steps

- Identify the planned duration
- Identify the actual duration
- Calculate the variance
- Express the variance as a percentage

430

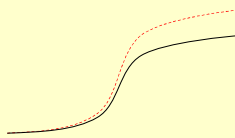
Schedule variance (SV) example

- For example, if a task was planned to take 10 days to complete but actually took 12 days, the schedule variance would be calculated as follows:
- $\text{Variance} = (12 - 10) / 10 = .2$
- $\text{Schedule variance (SV)} = 0.2 \times 100 = 20\%$
- This means that the task took 20% longer than planned

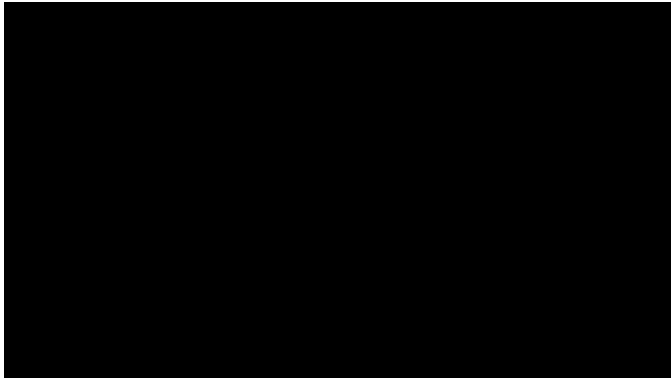
431

Schedule variance as a useful measure

- Identify tasks that are taking longer than expected
- Take corrective action to get a project back on track
- Cost variance
- Scope variance



432



433

Work breakdown structures (WBS)

434

Work breakdown structures (WBS)

- A hierarchical decomposition of a project into smaller, more manageable components
- A tool used for the total scope of a project
- Components are defined and organized
- Scope decomposed into manageable parts

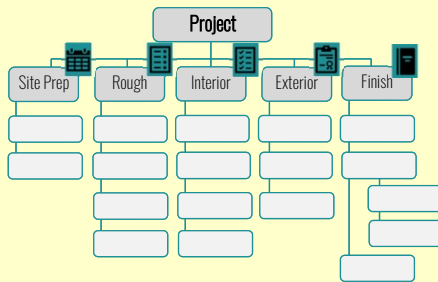
435

Work breakdown structures (WBS)

- Consists of a top-level summary of the project
- A series of increasingly detailed levels
- Components can be organized by deliverable, phase, other criteria

436

Work Breakdown Structure (WBS)



437

Benefits of the WBS

- Understand the scope of the project
- Identify the work that needs to be done
- Organize and track the progress of the project
- Ensure that all necessary tasks are included in the project plan

438

Example outline of a WBS

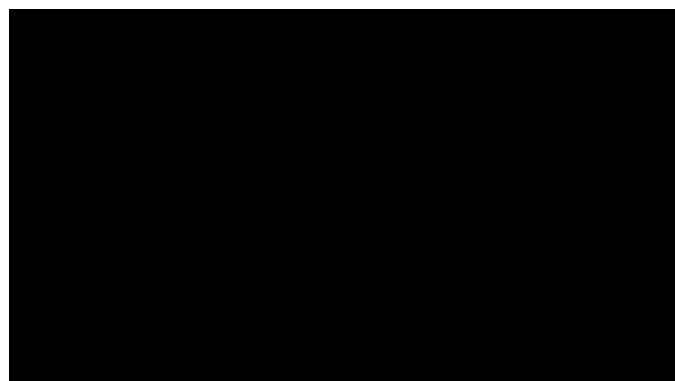
- Project summary
 - Phase 1
 - Deliverable 1
 - Deliverable 2
 - Phase 2
 - Deliverable 1
 - Deliverable 2
- Support tasks
- Project management

439

Project summary

- Objectives
- Deliverables
- Key stakeholders

440



441

Work package

442

Work package

- Represents a portion of the work to be completed to achieve the goals of the project
- Smallest item in WBS
- Deliverable
- Set of tasks

443

Work package organization

- High-level work packages
- Smaller, more specific sub-packages

444

Work packages defined

- Resources required to complete them
- Deliverables that will be produced as a result

445

Work package purpose

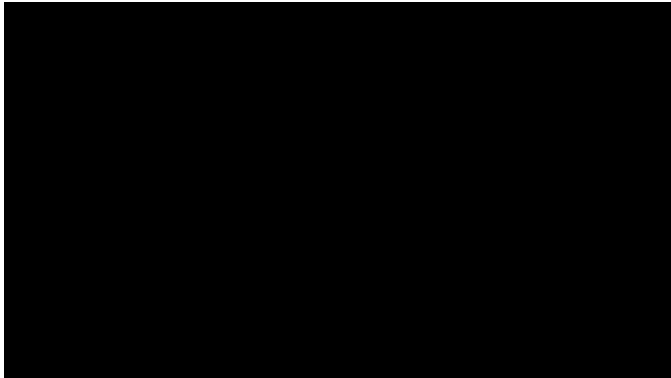
- Scope
- Requirements
- Project planning
- Progress tracking

446

Assigning work packages

- Specific team members
- Contractors
- Completing the tasks within the package
- Producing the associated deliverables

447



448

Quality management plan

449

Quality management plan

- Products
- Services
- Roles and responsibilities of all employees in maintaining quality
- Resources that will be used to achieve quality goals

450

Quality management plan

- Quality control
- Quality assurance
- Continuous improvement

451

Quality management plan

- Ensure effectiveness
- Make necessary updates
- Meet needs and expectations

452

Application of a quality management plan

- Review the project's objectives and requirements
- Identify the key stakeholders
- Define the quality standards
- Develop a quality control plan
- Implement the quality control plan
- Review and assess the quality management plan

453

Review objectives and requirements

- Goals
- Requirements
- Measured
- Validated

454

Identify key stakeholders

- Determine who the key stakeholders are for the project
- Needs and expectations

455

Define quality standards

- Determine standards used to evaluate the quality of the project's deliverables
- Industry standards
- Regulatory requirements
- Internal company standards

456

Develop a quality control plan

- Maintained
- Tested
- Identifying and addressing defects
- Ensuring the project meets defined quality standards

457

Implement the quality control plan

- Monitoring
- Testing
- QC inspection-driven

458

Review and assessment of the quality management plan

- Make any necessary changes
- Ensure it is meeting the needs of the project

459



460

Integration management plan

461

Integration management plan

- All areas of project affect one another
- Different systems or processes will be integrated
- The overall project will be coordinated
- Configuration management

462

Integration management plan

- Scope
- Cost
- Schedule
- Quality
- Resources
- Communication
- Risk
- Procurement
- Stakeholder engagement

463

Application of an integration management plan

- Determine the scope of the integration
- Develop an integration strategy
- Identify risks and issues
- Implement the plan
- Monitor and control progress
- Evaluate the results

464

Develop an integration strategy

- Bringing different parts together
- Coordinating activities
- Establishing communication channels
- Setting up shared resources
- Creating a system for tracking progress

465

Identify risks and issues

- Risk identification is an ongoing activity
- Status meetings
- Standup meetings/Daily Scrum
- Issues are risks that have occurred

466

Monitor and control progress

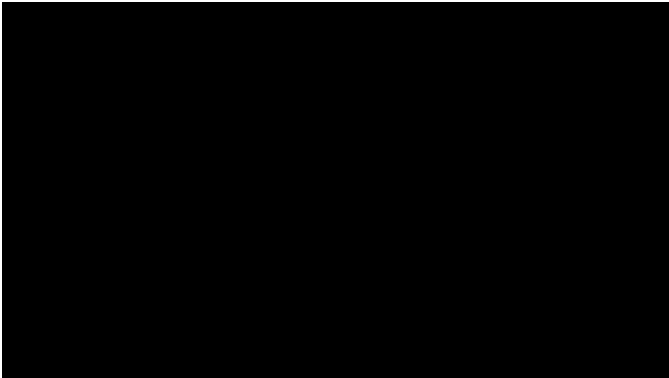
- Regularly review the progress of the integration process
- Make any necessary adjustments to the plan as needed
- Corrective action
- Preventive action

467

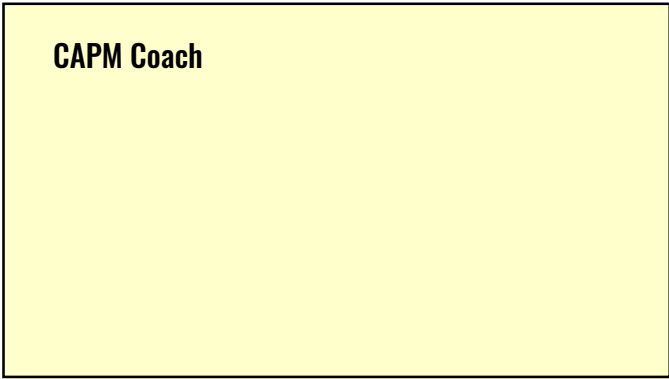
Evaluate the results

- Determine if the integration was successful
- Identify any areas for improvement

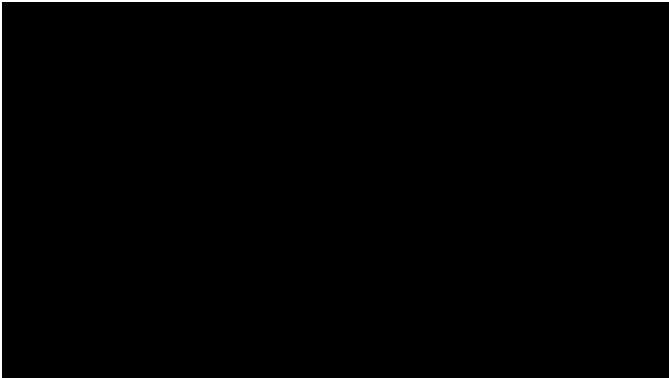
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469



470



471

Section conclusion

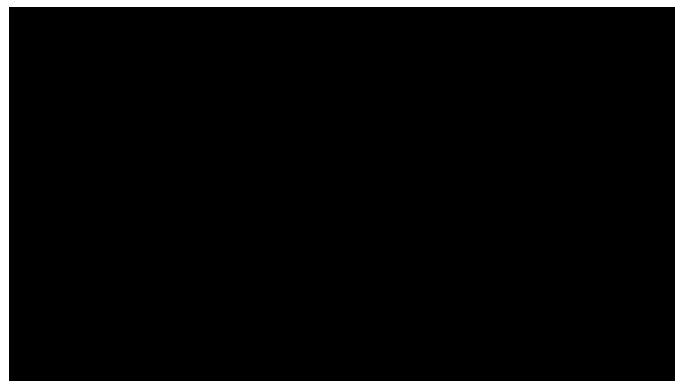
Demonstrate an understanding of a project management plan schedule

472

Section overview

Determine how to document project controls of predictive, plan-based projects

473



474

Project controls in predictive, plan-based projects

475

Project controls in predictive, plan-based projects

- Plan
- Monitor
- Control

476

Project controls in predictive, plan-based projects

- Stay on track
- Completed successfully – definition of done
- Completed within predetermined timeframe and budget

477

Project control activities

- Developing a detailed project schedule and budget
- Tracking progress and resource utilization against the plan
- Identifying and managing risks
- Making changes to the project plan as needed
- Generating regular reports on the progress of the project
- Reviewing and updating the project plan as the project progresses

478

Document project controls of predictive, plan based projects

- Establish project control procedures
- Document the project controls
- Use project management software
- Create reports
- Review and update the project plan

479

Establish project control procedures

- How progress will be monitored
- How changes will be managed
- How risks will be identified and addressed

480

Document the project control plan

- Create a written plan outlining the project control procedures
- Included in the overall project management plan
- Reviewed and updated regularly

481

Project management software

- Track progress
- Manage changes
- Create project schedules
- Track budget and resource utilization
- Identify potential risks and issues

482

Create reports

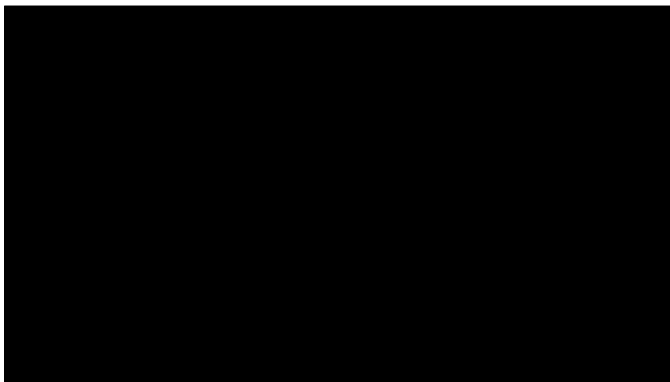
- Budget
- Schedule
- Quality
- Risks
- Issues
- Variances

483

Review and update the project plan

- This can be done as part of the overall project review process
- Ensure the plan is effective
- Allows for any necessary updates

484



485

Artifacts

486

Artifacts

- A document or other tangible item that is created or used during the project
- Artifacts can be used for all parts of project management

487

Artifacts

- Provide a record of the project's progress
- Decisions about how to best manage the project moving forward
- Organizational process asset
- Historical information

488

Artifacts used in predictive, plan-based projects

- Project charter
- Project plan
- Gantt chart
- Work breakdown structure (WBS)
- Risk register
- Change request log
- Status report
- Budget and cost management plan

489

Project charter

- Goals
- Stakeholders
- Key deliverables

490

Project plan

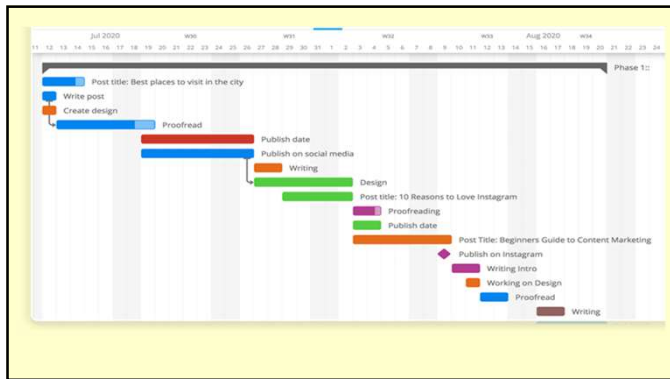
- Tasks
- Resources
- Timeline
- Goals/Objectives
- KPIs

491

Gantt chart

- Start and end dates of each task
- How tasks relate to one another

492



493

Risk register

- Identification
- Assessment
- Management

494

Change request log

- The reason for the change
- The impact on the project
- Any necessary approvals

495

Status report

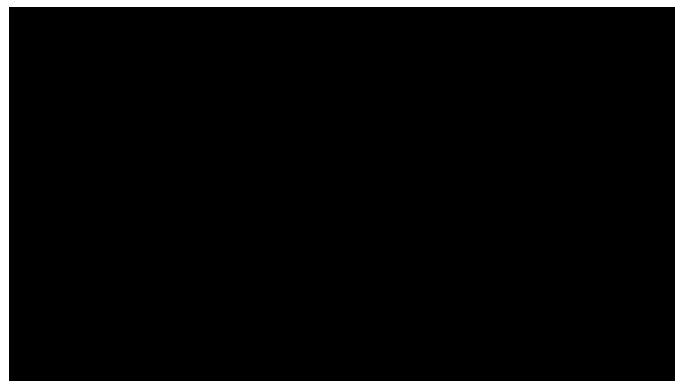
- Completed tasks
- Ongoing tasks
- Issues
- Challenges

496

Budget and cost management plan

- The project budget
- How costs will be tracked and managed

497



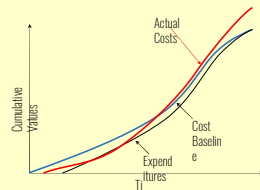
498

Cost variance

499

Cost variance

- A measure of how the actual cost of a project compares to the budgeted cost
- If the result is positive, then the project is over budget
- If the result is negative, then the project is under budget



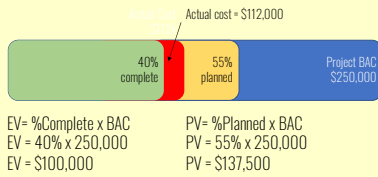
500

Measuring project performance

- Earned Value Management
- Forecast
- Measure performance
- Suite of formulas
- A few questions

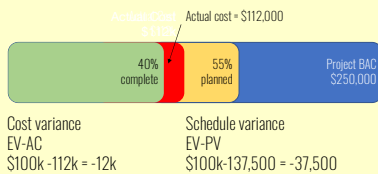
501

EVM foundation



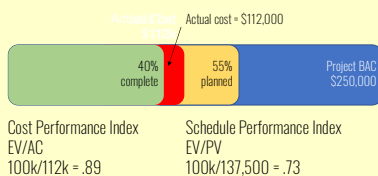
502

Finding the variances



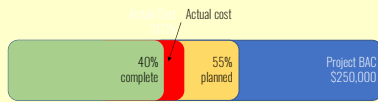
503

Measuring performance



504

Predicting the future



Estimate at Completion
BAC/CPI
 $\$250k / .89 = \$280,898$

Estimate to Complete
EAC-AC
 $\$280,898 - \$112k = \$168k$

Test questions will round up/down : answers and choices won't be close in results.

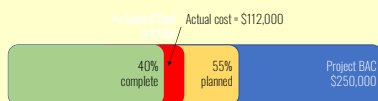
505

EAC: current cost performance

- Estimate at Completion (EAC)
- Future work at planned costs formula
- Forecasts final project costs based on current performance
- $AC + (BAC - EV)$

506

Predicting the future



Estimate at Completion
Current performance
 $AC + (BAC - EV)$
 $\$112k + (250k - 100k)$
 $\$262k$

Estimate at Completion
Standard formula
BAC/CPI
 $\$250k / .89 = \$280k$

507

EAC: SPI and CPI factors

- Estimate at Completion (EAC)
- Same efficiency rate for time and cost
- Can make SPI or CPI weighted
- $EAC = AC + [(BAC - EV) / (CPI \times SPI)]$

508

Predicting the future



Estimate at Completion
Standard formula
 BAC / CPI
 $\$250k / .89 = \$280,898$

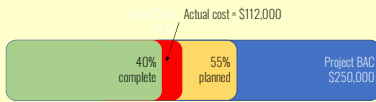
509

To-Complete Performance Index

- Can you meet the BAC?
- Can you meet the EAC?
- $TCPI = (BAC - EV) / (BAC - AC)$
- $TCPI = (BAC - EV) / (EAC - AC)$

510

$$TCPI = (BAC - EV) / (BAC - AC)$$



$$TCPI = (BAC - EV) / (BAC - AC)$$

$$TCPI = (250k - 100k) / (250k - 112k)$$

$$TCPI = (150k) / (138k)$$

$$TCPI = 1.09$$

511

$$TCPI = (BAC - EV) / (EAC - AC)$$



$$TCPI = (BAC - EV) / (EAC - AC)$$

$$TCPI = (250k - 100k) / (280k - 112k)$$

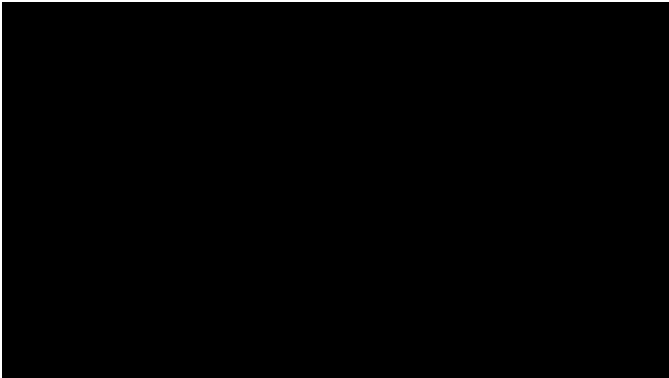
$$TCPI = (150k) / (168k)$$

$$TCPI = .89$$

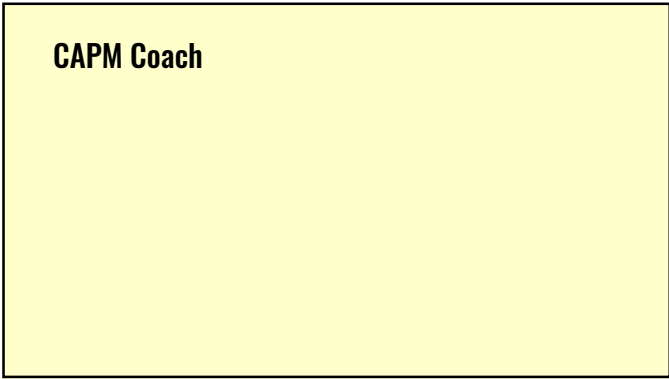
512

EVW formula worksheet

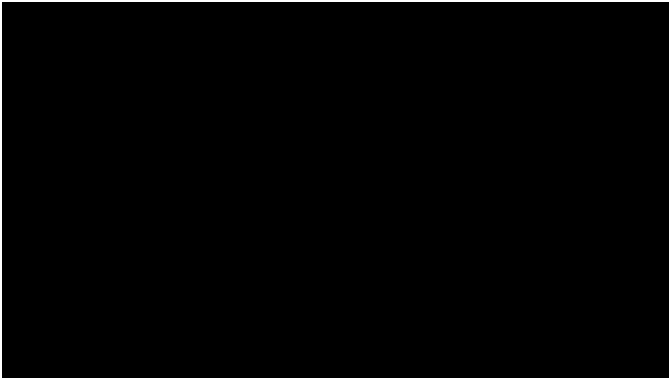
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514



515



516

Section conclusion

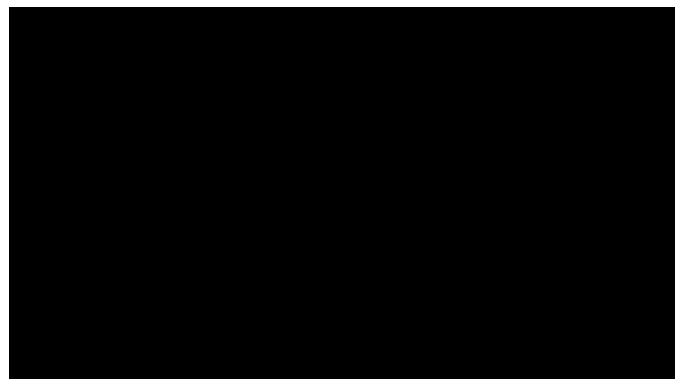
Determine how to document project controls of predictive, plan-based projects

517

Section overview

Explain when it is appropriate to use an adaptive approach

518



519

Adaptive approaches

520

Adaptive approaches

- Changes or variations in the environment or inputs
- Changing conditions
- Varying needs

521

Appropriate use of adaptive approaches

- Environment is uncertain or dynamic
- Important to optimize performance
- Changing requirements or constraints

522

Uncertain or dynamic environments

- Frequent
- Significant
- Flexible
- Responsive
- Fluctuating demand
- Changing customer needs

523

Optimizing performance

- System
- Process
- Adjust
- Fine-tune itself

524

Adapting to changing requirements or constraints

- System
- Process
- Adapting to new regulations or standards
- Adapting to changes in the availability or cost of resources

525

Adaptive vs predictive, plan-based projects

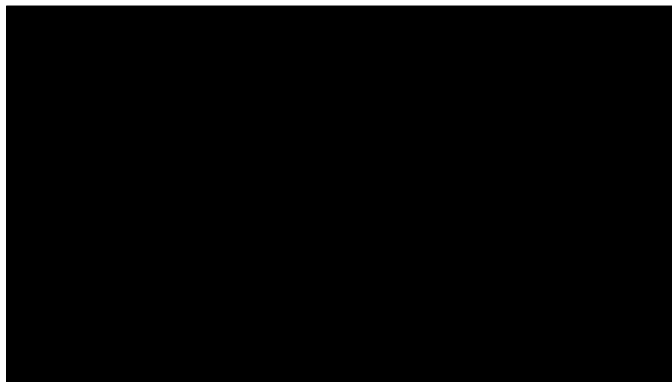
Adaptive

- An iterative process of planning, executing and reviewing progress
- A focus on adapting to changes and uncertainties as they arise

Predictive

- Creating a detailed plan up front
- Following the plan as closely as possible

526



527

Pros and cons of adaptive projects

528

Pros and cons of adaptive projects

Pros

- Flexibility
- Greater agility
- Improved decision making

Cons

- Lack of upfront planning
- Greater risk

529

Pros and cons of predictive plan-based projects

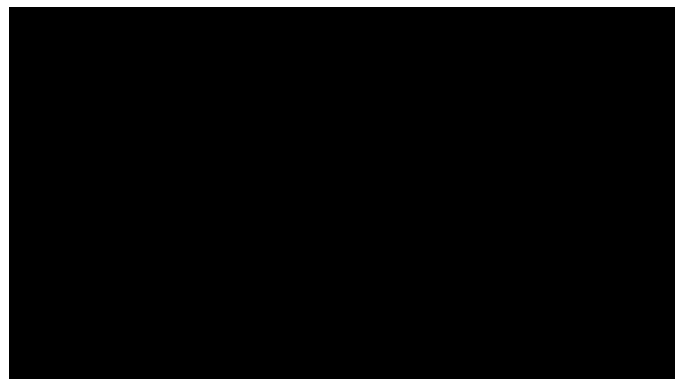
Pros

- Greater predictability
- Improved risk management

Cons

- Inflexibility
- Greater risk of failure

530



531

Adaptive approaches for the organizational structure

532

Adaptive approaches for the organizational structure

- Planning
- Executing
- Reviewing
- Changes
- Uncertainties

533

Virtual structure

- An adaptive approach may be particularly suitable for virtual teams
- Can be helpful in situations where team members are geographically dispersed
- Allows for greater flexibility
- Ability to adapt to changes as they occur

534

Collocated structure

- An adaptive approach may be suitable for collocated teams
- Can be helpful in quickly responding to changes or challenges
- Communication and collaboration
- Osmotic communication

535

Matrix structure

- May be suitable for teams operating in matrix structure
- Coordination and collaboration
- The ability to adapt to changes as they arise

536

Hierarchical structure

- An adaptive approach may be less suitable for teams operating in a hierarchical structure
- May be more difficult to quickly make changes and adjustments
- Waiting on approval rather than quick decisions
- Waiting time is waste

537

Osmotic communication

- A benefit of collocated teams is osmotic communication
- Useful information that flows between team members who are in proximity

538

Distributed teams

- At least one team member is working off-site
- Majority of agile teams are distributed
- Communication tools and reduced communication costs
- Cost-effective approach

539

Global, cultural, and team diversity

- Not uncommon to have team member from different countries working together
- May create challenges:
 - Time zones
 - Cultures
 - Communication styles
 - Native languages

540

Digital tools for distributed teams

- Videoconferencing, live chat, Skype
- Interactive whiteboards
- Instant messaging
- Web collaboration software

541

Digital tools for distributed teams

- Presence-based applications
- Electronic task boards and story boards
- Web-based meeting facilitators
- Survey applications
- Agile project management software
- Virtual card walls
- Monday.com, Trello, Wrike, Smartsheet

542

Distributed vs. outsourced teams

- Distributed team: multiple development sites that can span buildings, cities, or countries
- Outsourced team: multiple legal entities, contracting, contract administration and different development infrastructures

543

Encourage emergent leadership

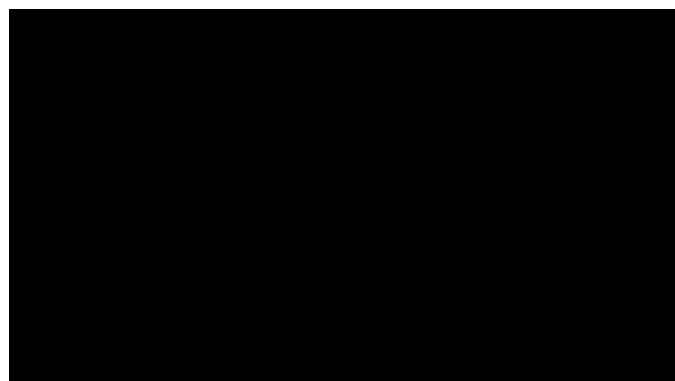
- Team success is more important than individual success
- Anyone can be a leader on an agile team
- Understanding roles and responsibilities

544

Fostering emergent leadership

- Allow team members to take charge
- Motivate the team members
- Show recognition of individuals and team
- Avoid zero sum rewards

545



546

Factors influencing suitability of adaptive approaches for organizational structures

547

Level of uncertainty

- Uncertain
- Dynamic
- Greater flexibility
- Ability to adapt to changes as they occur

548

Organization culture

- Attitude towards framework
- Changes
- Challenges

549

Size and complexity of projects

- Smaller or less complex projects require less upfront planning
- Flexible in responding to changes

550

Level of team autonomy

- Involves regular reviews
- Evaluation of progress
- Ability to adjust the plan as needed

551

Stakeholder preference

- Stakeholders' preferences and needs may impact the approach
- Resource constraints
- Risk tolerance
- Project goals and priorities
- Level of influence

552

Organizational process assets

- Resources
- Processes
- Tools
- Historical information
- Artifacts

553

Environmental factors

- Enterprise and external factors
- Conditions
- Influences

554

Organizational process assets that facilitate the use of adaptive approaches

- Project management software
- Risk management processes
- Agile methodologies
- Collaboration tools

555

Project management software

- Planning
- Tracking progress
- Adapting to changes

556

Risk management processes

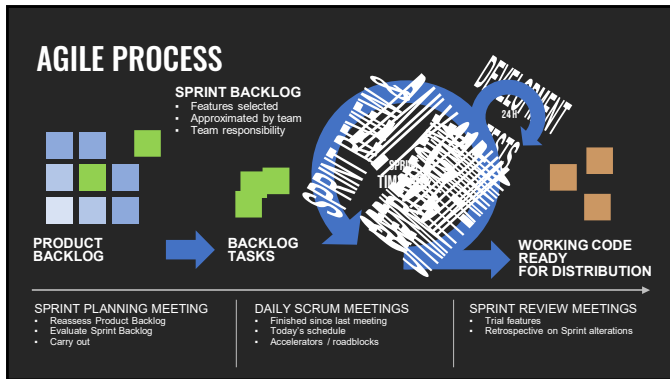
- Help to identify potential risks or challenges
- Plan for potential risks or challenges

557

Agile methodologies

- Scrum
- Lean
- Kanban

558

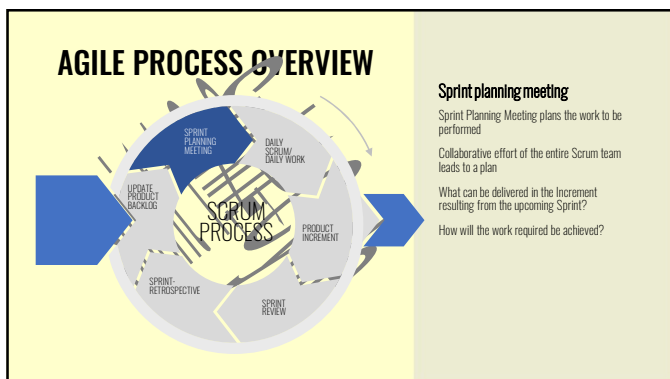


559

Pre-sprint activities

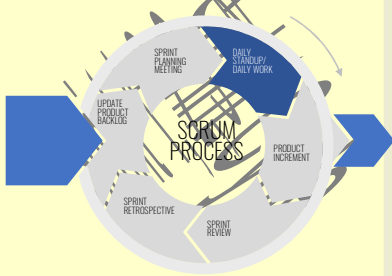
- Vision Statement: concise description of the goals of the project
- Product Roadmap: visual timeline of major product features to be delivered and is normally created by the Product Owner
- Stories: requirements written by the Product Owner and based on customer requirements
- Product Backlog: made up of stories and prioritized

560



561

AGILE PROCESS OVERVIEW



The diagram illustrates the Scrum process as a continuous cycle. It starts with a 'PRODUCT BACKLOG' (represented by a blue arrow) entering a central 'SCRUM PROCESS' circle. The cycle includes: 'SPRINT PLANNING MEETING' (top), 'DAILY SCRUM/DAILY WORK' (top-right), 'PRODUCT INCREMENT' (right), 'SPRINT REVIEW' (bottom-right), 'SPRINT RETROSPECTIVE' (bottom), and 'UPDATE PRODUCT BACKLOG' (bottom-left). Arrows indicate the flow from one stage to the next, eventually leading to a 'PRODUCT INCREMENT' (represented by a blue arrow) exiting the cycle.

Daily Scrum

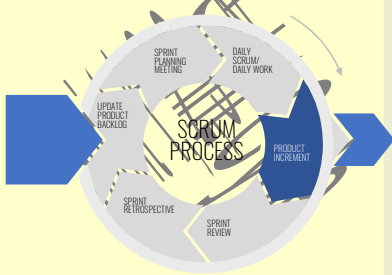
15-minute event
Development Team synchronizes activities and create a daily plan

Three questions:

1. What did I do yesterday that helped the Development Team meet the Sprint Goal?
2. What will I do today to help the Development Team meet the Sprint Goal?
3. Do I see any stumbling blocks that prevent me or the team from meeting the Sprint Goal?

562

AGILE PROCESS OVERVIEW



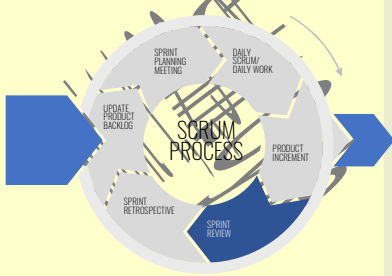
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Product Increment / finished work

Increment is the sum of all the Product Backlog items completed during a Sprint.
Includes the Increments of all previous Sprints.
At the end of a Sprint, the new Increment must be "Done"
The product must be useable regardless of whether the Product Owner decides to release it

563

AGILE PROCESS OVERVIEW

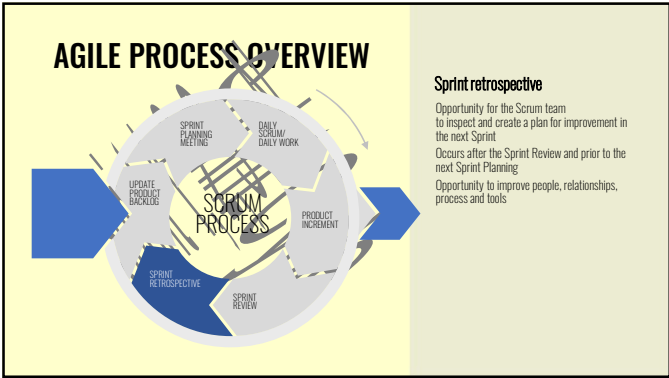


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Sprint review

Sprint Review at the end of Sprint
Inspect the Increment and may adapt the Product Backlog
Scrum team and stakeholders collaborate about what was done in the Sprint
Informal meeting, not a status meeting

564



565

Cancelling a sprint

- Product Owner has the authority to cancel a sprint
- Can happen when the Sprint Goal becomes obsolete
 - Changes in the Product Backlog, strategies, or approach
- When a sprint is cancelled, the items that are "Done" will be reviewed and accepted, and the rest of the items will be put back into the Product Backlog

566

Collaboration tools

- Collaboration tools are essential for an adaptive approach
- Communication
- Collaboration
- Team communication
- Project management software

567

Environmental factors that facilitate the use of adaptive approaches

- Dynamic or uncertain environments
- Stakeholder preferences
- Regulatory or compliance requirements
- Technological changes

568

Dynamic or uncertain environment

- Greater flexibility
- The ability to adapt to changes as they occur

569

Regulatory compliance requirements

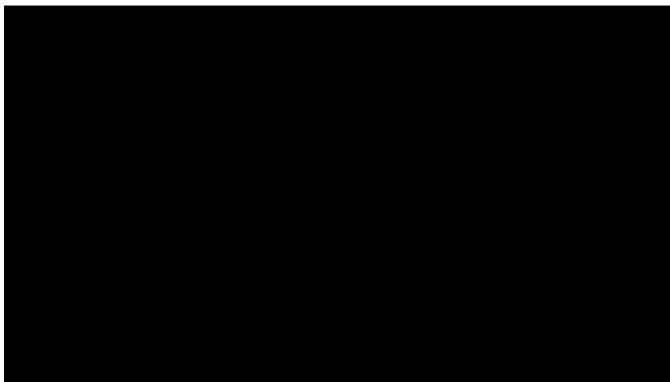
- An adaptive approach may be more suitable for ensuring that the project remains compliant if regulatory or compliance requirements are subject to frequent changes

570

Technological changes

- Rapid technological changes can make an adaptive approach more suitable
- Adaptive approaches allow for the incorporation of new technologies as they become available

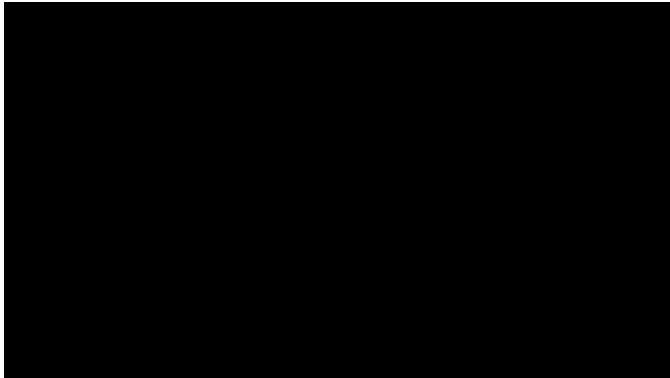
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572

CAPM Coach

573



574

Section conclusion

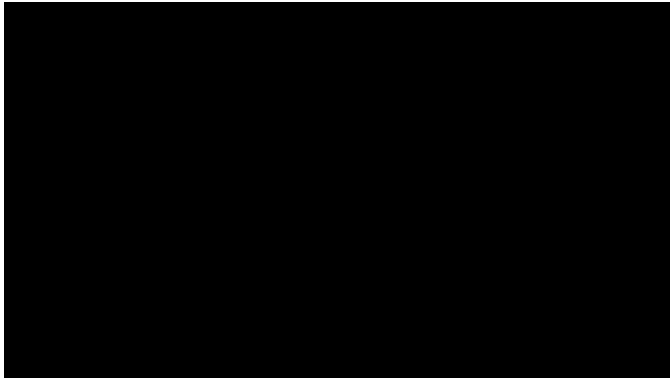
Explain when it is appropriate to use an adaptive approach

575

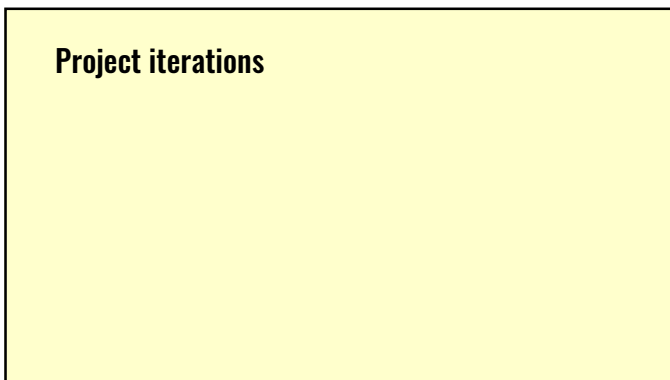
Section overview

Determine how to plan project iterations

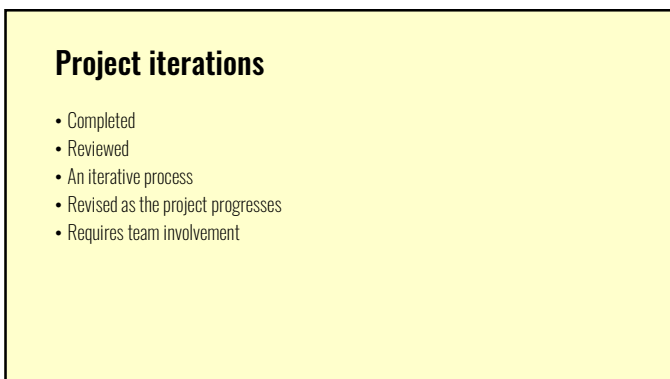
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577



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579

How to plan project iterations

- Define project goals and objectives
- Break the project down into small, manageable chunks of work
- Determine iteration lengths - one week to one month
- Set clear and measurable goals
- Define the scope
- Create a schedule
- Identify risks or issues
- Review and adjust as needed

580

User Story: Card

User Story Title
As a <user> I want to <goal> so that <benefit>.

TEMPLATE

Enter sales in an application
As a salesperson
I want to enter sales on a phone app so customers receive products faster.

581

Logical units of iterations

- Iteration goal
- Iteration scope
- Iteration schedule
- Iteration review

582

Iteration scope

- Small enough to be achievable
- Large enough to be meaningful and valuable
- Example: design the login page, implement the login functionality, and write unit tests

583

Iteration schedule

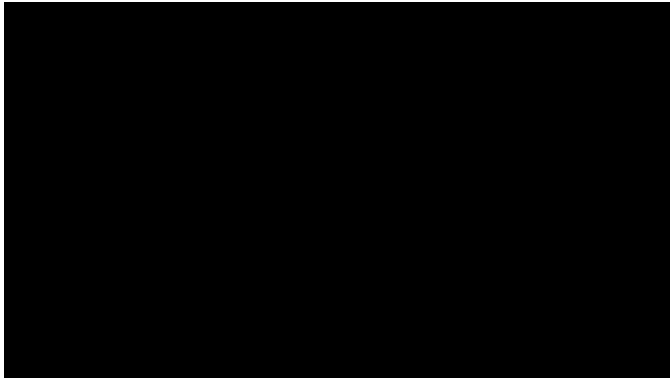
- Tasks that need to be done
- Resources that will be needed
- Dependencies between tasks
- Team: self-led and self-organizing

584

Iteration review

- Team demonstrates completed work for iteration
- Assess the team progress
- Identifies any issues or risks

585



586

Interpret pros and cons of the iteration

587

Interpret pros and cons of the iteration

- Needs
- Goals
- Carefully consider suitability
- Choose the approach that best supports project success

588

Pros and cons of the iteration

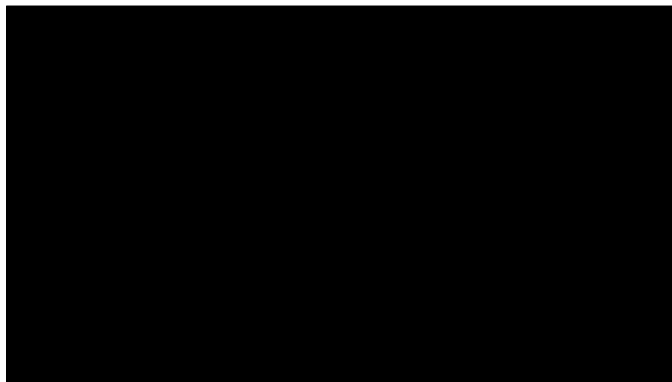
Pros

- Allows for flexibility and adaptability
- Promotes collaboration and communication
- Facilitates risk management
- Increases productivity

Cons

- Can be more time-consuming
- Requires more planning
- May not be suitable for all projects

589



590

Work breakdown structure (WBS)

- Smaller
- Manageable
- Define
- Organize

591

Translating a WBS to an adaptive iteration

- Identify the scope of the project
- Break down the scope into smaller, more manageable chunks
- Organize the work packages into a hierarchy
- Determine the dependencies between the work packages
- Estimate the duration and resources needed for each work package
- Monitor and control the progress of the work packages
- Adapt to changing circumstances as needed

592

Estimate duration and resources

- Create a project schedule
- Allocate necessary resources
- Story point sizing
- Fibonacci sequence: 1, 3, 5, 8, 13
- Planning poker

593

Monitor and control progress

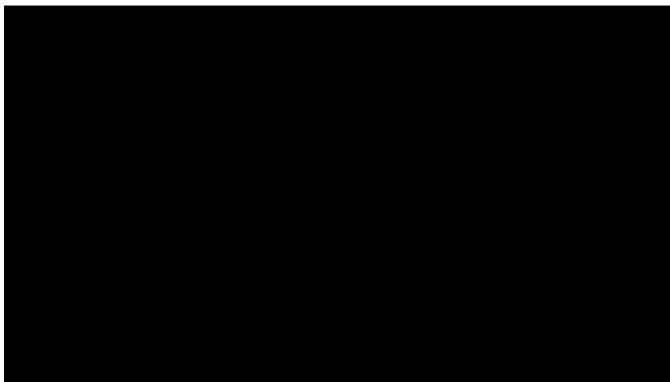
- On-time completion
- Completion within budget

594

Adapt and change

- Unexpected challenges
- Opportunities
- Flexible
- Open
- Adaptable

595



596

Inputs for project scope

597

Inputs for project scope

- Project charter
- Business case
- Customer requirements
- Market research
- Regulatory requirements
- Organizational constraints

598

Project charter

- Purpose
- Objectives
- Stakeholders
- Scope of the project

599

Business case

- How the project fits into the overall business strategy
- Justification for the project
- Potential project benefits
- Project costs

600

Customer requirements

- Needs
- Expectations
- Customer surveys
- Focus groups
- Other methods

601

Market research

- Help to inform the scope of the project
- Ensure project scope meets the needs and expectations of the intended audience

602

Requirements

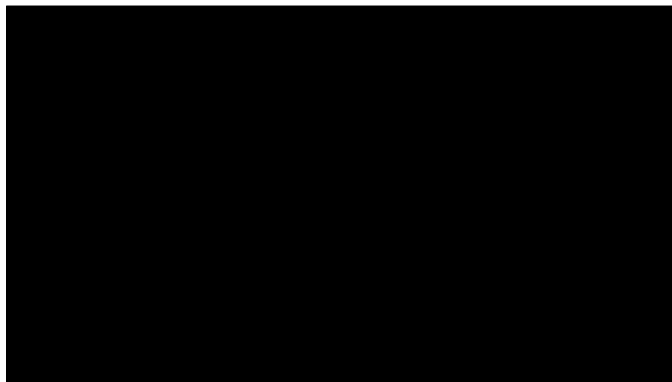
- Legal
- Regulatory
- Understand requirements
- Ensure compliance

603

Organizational constraints

- Resources and capabilities
- Defining the scope
- Making sure that the project is feasible
- Ensuring the project aligns with the organization's goals

604



605

Adaptive project tracking

606

Adaptive project tracking

- Agility
- Flexibility
- Plans
- Activities
- Learning
- Feedback

607

Predictive, plan-based tracking

- Creating a detailed project plan at the outset of the project
- Tracking progress against that plan

608

Important benefits of adaptive planning

- Improved accuracy
- Enhanced agility
- Greater stakeholder engagement
- Better risk management

609

Improved accuracy

- Forecasting
- Estimation
- New information
- Changing circumstances

610

Greater stakeholder engagement

- Communication
- Collaboration
- Higher levels of engagement
- Buy-in

611

Better risk management

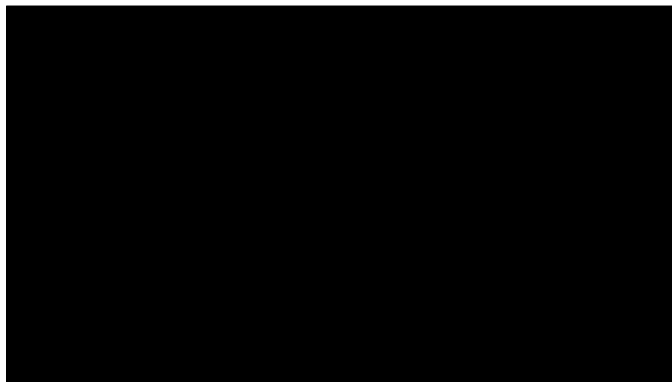
- Mitigate risks
- Address issues

612

Adaptive project tracking versus predictive, plan-based tracking

- Complex
- Uncertain
- Changing

613



614

CAPM Coach

615



616

Section conclusion

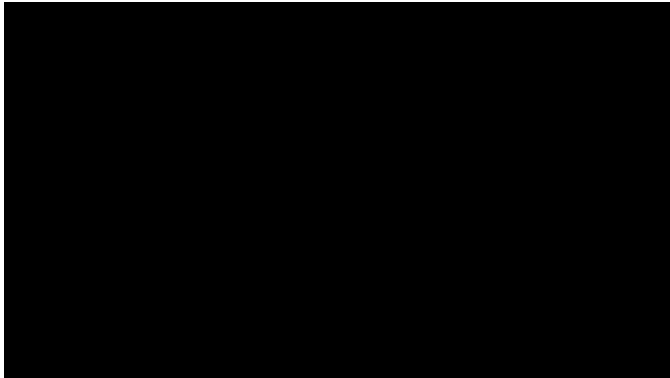
Determine how to plan project iterations

617

Section overview

**Determine how to document project controls
for an adaptive project**

618



619

Documenting project controls in adaptive projects

620

Documenting project controls in adaptive projects

- Identify the key project controls
- Create a system for tracking and monitoring
- Document the project controls in a central location
- Update the project controls regularly
- Use the project controls to inform decision-making
- Communicate the project controls to stakeholders

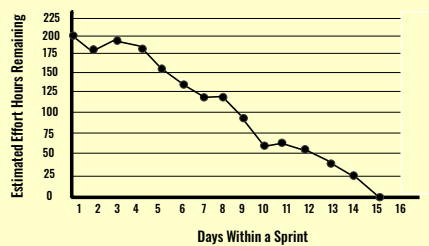
621

Create a system for tracking and monitoring

- Information radiator
- Daily Scrum
- Story points
- Velocity

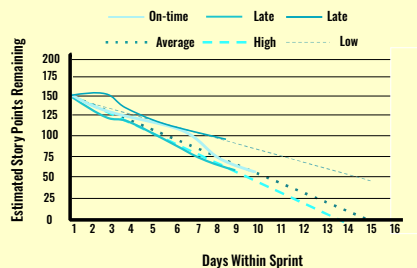
622

Sprint burndown chart



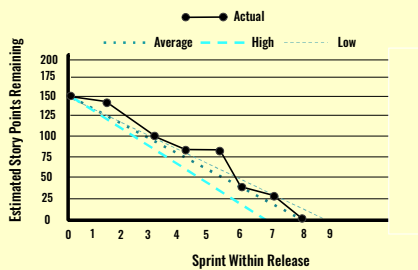
623

Sprint burndown chart with trend lines



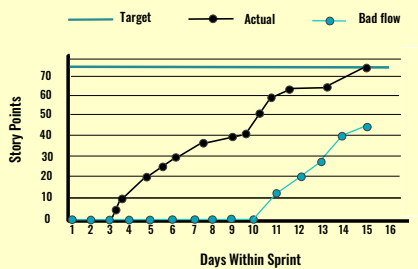
624

Fixed-scope-release burndown chart



625

Sprint burnup chart



626

Update project controls regularly

- Project plan
- Risks
- Other key factors

627

Use project controls to inform decision-making

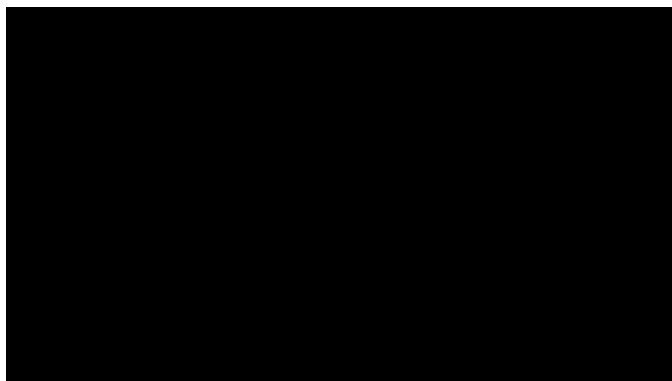
- Guide decision-making
- Help the team stay aligned with the project goals

628

Communicate project goals to stakeholders

- Ensure that everyone is working towards the same goals
- Project controls
- Build trust

629



630

Artifacts used in adaptive projects

631

User stories

- Short
- Simple
- Capture project requirements
- Help the development team understand the users' needs

632

Product roadmap

- Key features
- Milestones
- Prioritize
- Plan

633

Iteration plan

- Specific tasks
- Resources

634

Sprint backlog

- A list of all the work that needs to be completed in a specific sprint
- User stories
- Other tasks that need to be done
- Estimates of how long tasks will take

635

Burn down chart

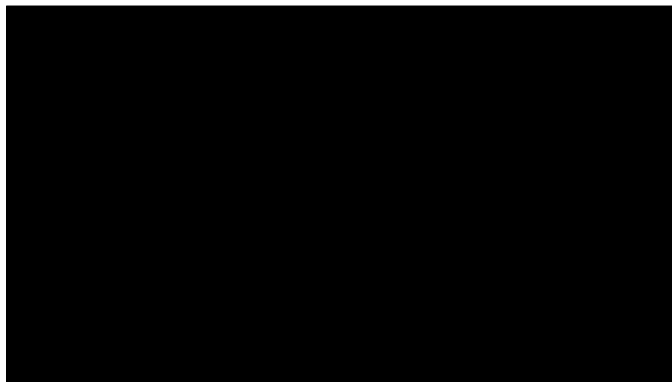
- The amount of work remaining in the project over time
- Team progress
- Potential issues

636

Retrospective

- Review the work that was completed
- Identify any areas for improvement
- Continuously improve their processes
- Deliver better results

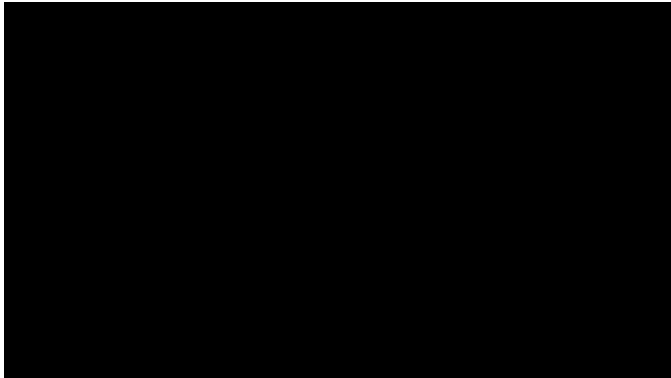
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638

CAPM Coach

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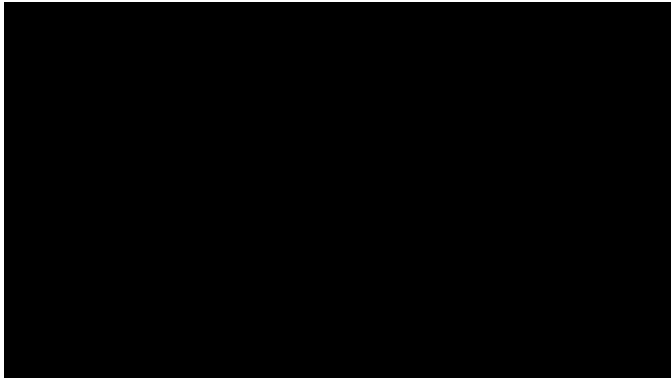
Section conclusion

641

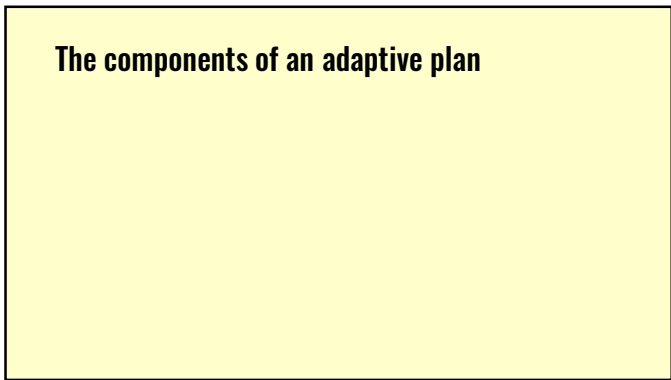
Section overview

Explain the components of an adaptive plan

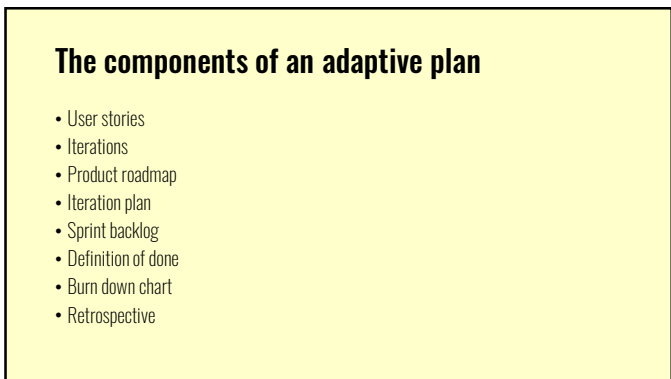
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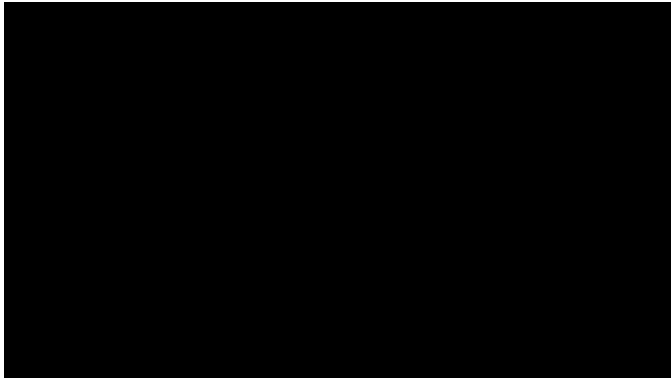
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645



646

Components of different adaptive methodologies

647

Scrum framework

- Scrum Master
- Product Owner
- Development team
- Sprint planning
- Daily Scrum
- Sprint reviews
- Retrospectives

648

eXtreme programming (XP) framework

- Iterations/increments
- Pair programming
- Unit testing/acceptance testing
- Business logic – understand the customer needs
- Communication and simplicity

649

Lean framework

- Eliminate waste
- Amplify learning
- Decide as late as possible
- Deliver as fast as possible
- Empower the team
- Build integrity in (QA)
- See the whole

650

SAFe framework

- Scaled Agile Framework
- Agile principles for large organizations
- Business alignment
- Transparency
- Systems thinking
- Incremental builds
- Organize around value

651

Crystal framework

- Frequent delivery
- Reflective improvement
- Osmotic communication
- Personal safety
- Focus
- Easy access to expert users
- Technical tooling

652

Dynamic systems development method framework

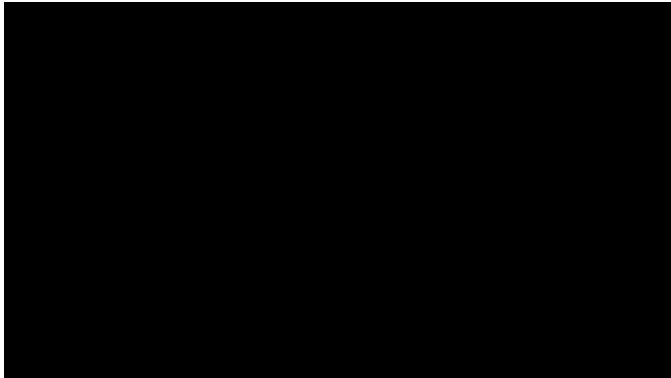
- MoSCoW: must, should, could, will not
- Focus on the business need
- Deliver on time
- Collaborate
- Never compromise quality
- Build incrementally from firm foundations
- Develop iteratively
- Communicate continuously and clearly
- Demonstrate control

653

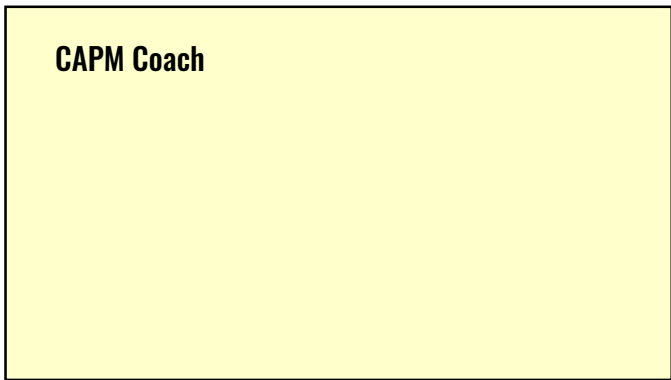
Kanban

- Track the status of work items
- Identify bottlenecks in the process
- Kanban board (sign board)

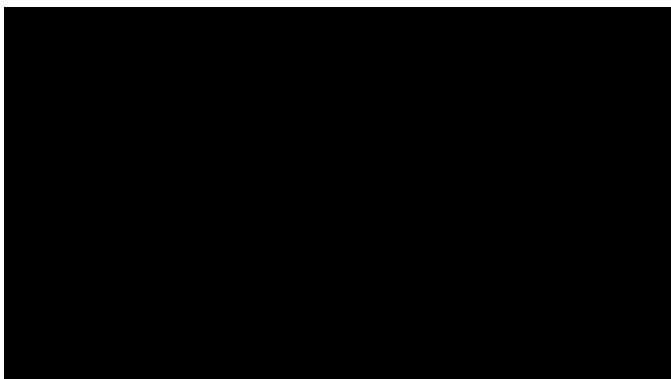
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Section conclusion

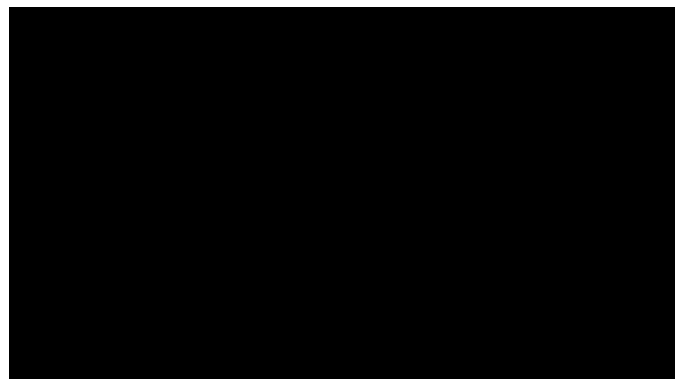
Determine how to document project controls for an adaptive project

658

Section overview

Determine how to prepare and execute task management steps

659



660

How to prepare and execute task management steps

661

Identify the task

- What needs to be done
- What the outcome should be

662

Estimate the effort

- How long the task is likely to take
- The resources that will be needed

663

Prioritize the task

- Determine the importance of the task relative to other tasks
- Assign the task a priority level

664

Assign the task

- Decide who will be responsible for completing the task
- Assign the task to the responsible person

665

Break the task down

- Small
- Manageable

666

Create a schedule

- Schedule
- Timeline

667

Track progress

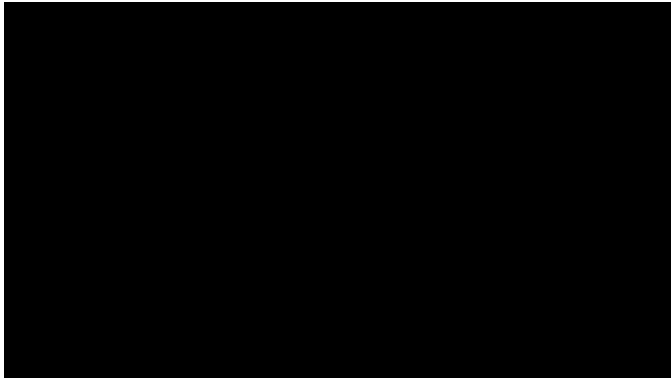
- The task is on track
- Make any necessary adjustments

668

Review and adjust

- Review the results
- Make any necessary adjustments to the process

669



670

Success criteria in adaptive project management tasks

671

Success criteria in adaptive project management tasks

- Progress
- Success – Definition of Done (DoD)
- Clear
- Measurable
- Achievable
- Aligned with the overall objectives of the project

672

Success criteria examples

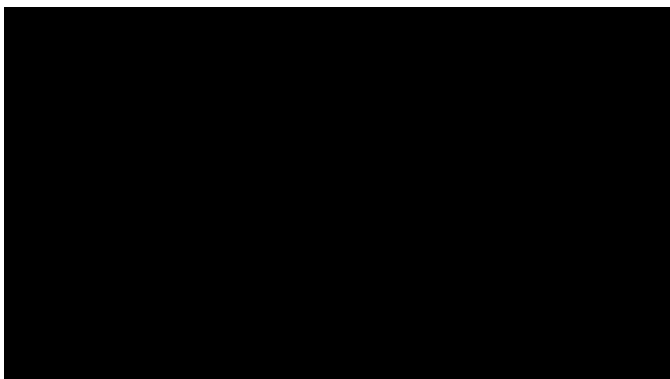
- Meeting deadlines for deliverables
- Staying within budget
- Achieving a certain level of quality in the final product
- Meeting or exceeding customer or stakeholder expectations
- Improving efficiency or productivity
- Meeting regulatory or compliance requirements

673

Establishing success criteria

- Guide decision-making
- Ensure the project stays on track
- Identify issues
- Pinpoint challenges need to be addressed

674



675

Prioritizing tasks in adaptive project management

676

Prioritizing tasks in adaptive project management

- Most important
- Time-sensitive
- Urgency
- Importance
- Dependencies
- Resources
- Risk

677

Urgency

- Have a tight deadline
- Need to be completed in order to meet a critical milestone

678

Importance

- Highest priority should be given to tasks that are critical to the success of the project

679

Dependencies

- Tasks that depend on the completion of other tasks should be given priority
- Prioritizing dependencies helps ensure the project stays on track

680

Resources

- Scarce
- Specialized
- Prioritizing resources will help to ensure they are used effectively

681

Risk

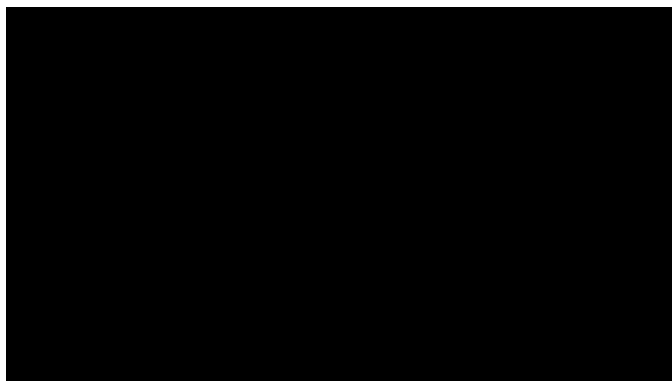
- Tasks that carry a high level of risk should be given priority
- Prioritizing risk minimizes the potential impact on the project

682

Prioritizing tasks in adaptive project management

- Create a list of all the tasks that need to be completed
- Rank the tasks according to the criteria
- Ensure the team is focused on the most important work
- Prevent delays or roadblocks

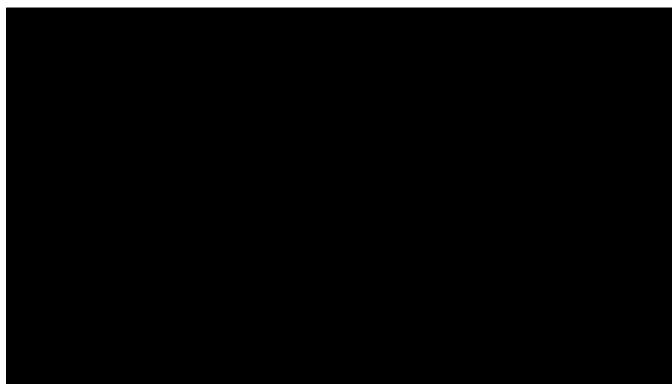
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CAPM Coach

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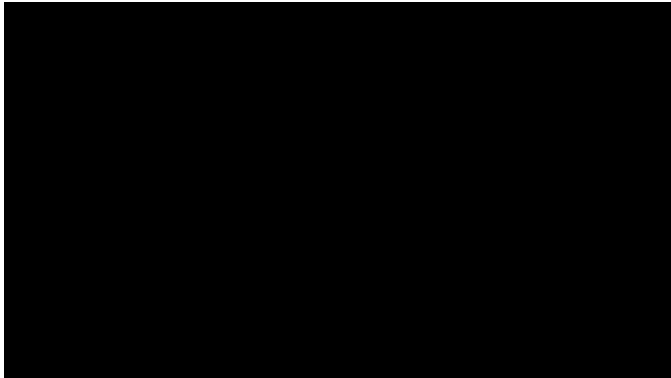


686

Section conclusion

Determine how to prepare and execute task management steps

687

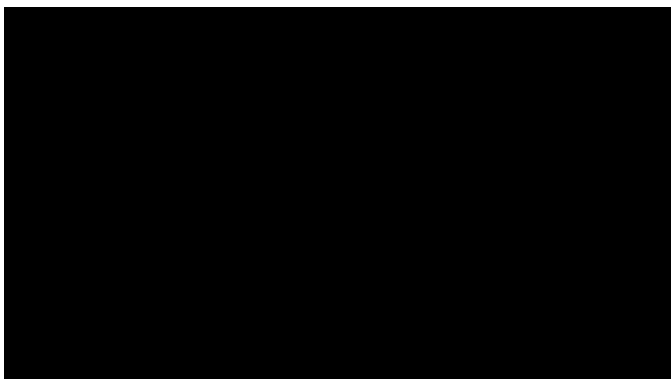


688

Section overview

Demonstrate an understanding of business analysis (BA) roles and responsibilities

689



690

The role of a business analyst in adaptive project management

691

The role of a business analyst in adaptive project management

- What is adaptive project management?
- What is the role of the BA?
- How does the BA contribute to the success of a project?

692

Adaptive project management

- Flexible approach
- Allows for changes and adjustments
- Requirements are uncertain
- Projects are complex

693

Role of the business analyst

- Processes
- Products
- Services
- Business needs
- Recommend solutions

694

Responsibilities of the business analyst

- Gathering and analyzing data
- Identifying problems and opportunities
- Developing recommendations and solutions
- Communicating with stakeholders

695

The BA's contribution to the success of the project

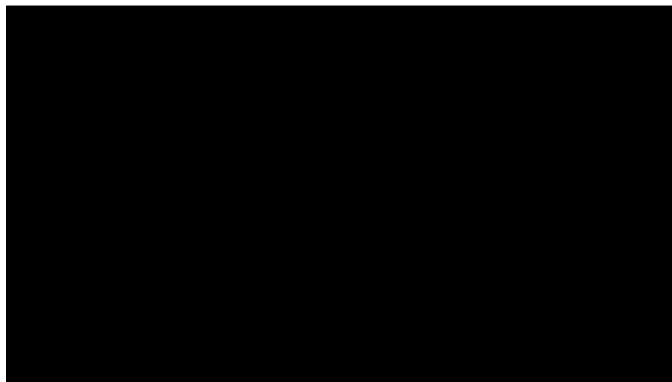
- Identifies areas for improvement
- Allows the team to proactively address issues
- Develops solutions to address identified problems or opportunities
- Serves as a key point of communication between the team and stakeholders

696

Benefits of a business analyst in adaptive project management

- Improved decision making
- Enhanced stakeholder communication
- Increased efficiency

697



698

Distinguish between stakeholder roles

699

Process owner

- Overall design
- Implementation
- Maintenance
- Defining the process
- Establishing performance standards
- Ensuring that the process is followed consistently

700

Process manager

- Responsible for the day-to-day management of the process
- Monitoring performance
- Identifying and addressing problems
- Ensuring that the process is operating efficiently

701

Product manager

- Overall strategy
- Direction
- Defining the product roadmap
- Setting goals
- Working with cross-functional teams
- Ensuring the product meets customer and stakeholder needs

702

Product Owner

- Responsible for representing stakeholder interest in product development
- Prioritizing features and functionality
- Defining acceptance criteria
- Working with development teams to ensure that products meet user needs

703

Additional stakeholders

- Project manager
 - Responsible for overall project coordination and management
- Management
 - Responsible for overseeing the project
- Project team
 - Responsible for executing the work of the project

704

Additional stakeholders

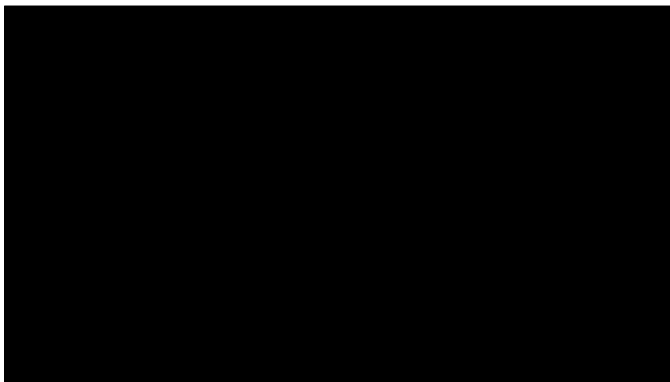
- Business analyst
 - Responsible for analyzing business needs and identifying opportunities for improvement
- User experience designer (UX)
 - Responsible for creating designs that are intuitive and easy to use for users
- Customers
 - People who will use the product or service being developed

705

Additional stakeholders

- Sponsors
 - Individuals or groups providing the funding or resources
 - Departments within the organization
- External stakeholders
 - Individuals or groups outside of the organization affected by the project

706



707

Outline the need for roles and responsibilities

708

Outline the need for roles and responsibilities

- Improve communication
- Enhance accountability
- Create greater efficiency
- Allow for better decision-making

709

Improved communication

- Less confusion
- Less likelihood of duplicated or missed tasks
- Improved coordination with the project team

710

Enhanced accountability

- Ensures timely task completion
- High-quality standards are likely to be met

711

Influences on resource management

- Team environment and politics
- Communication demands
- Organizational change management
- Culture and organization issues
- Challenges of virtual teams and geographic concerns

712

Resource terminology

- Role is the generic project team name
 - Application developer
 - Technical writer
- Authority is the level of decision-making ability
- Responsibility are actions and expectations of work to complete
- Competency is the role's depth of skills, knowledge, and experience

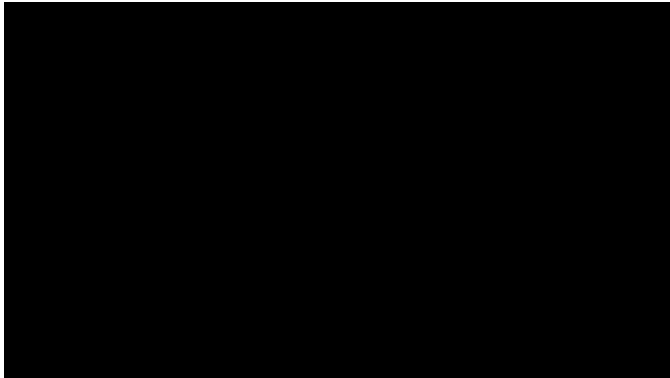
713

Matrix chart

RACI Chart
Responsibility Assignment Matrix
Roles and Responsibilities

Activity	Team Member					
	Sam	Shelly	Ben	Frank	Lloyd	Mark
Web content	R	A		C	I	
Web design	A	R				
App development	I		A		R	
Security	I		R	I	I	A
Proofing				A		
Testing				R		
Payment system	I		I	I	I	R

714



715

Differentiate between internal and external roles

716

The importance of internal and external roles

- Identification and engagement with both internal and external stakeholders
- Encourages different perspectives
- Generates various skillsets
- Produces resources
- Cultivates collaboration
- Fosters communication

717

Internal roles

- Project manager
- Project team
- Management
- Other departments

718

Project manager

- Coordination of the project
- Management of the project

719

Project team

- Responsible for executing the work of the project
- Developers
- Designers
- Testers
- Other specialists

720

Management

- Overseeing the project
- Ensuring the project aligns with the goals and objectives of the organization

721

Other departments

- May be impacted by the project
- May be involved in the project in some way

722

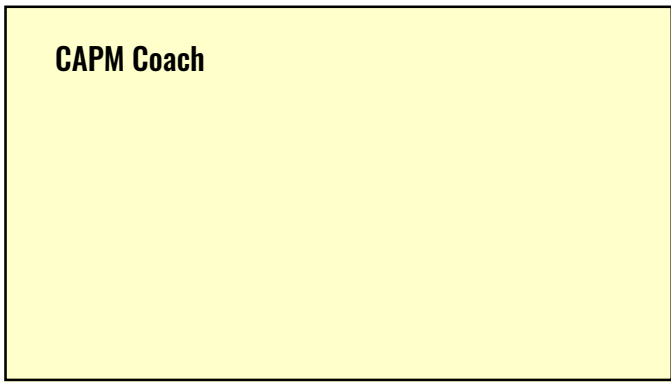
External roles

- Vendors
- Government agencies
- Contractors
- Competition
- Customers
- Unions
- Communities

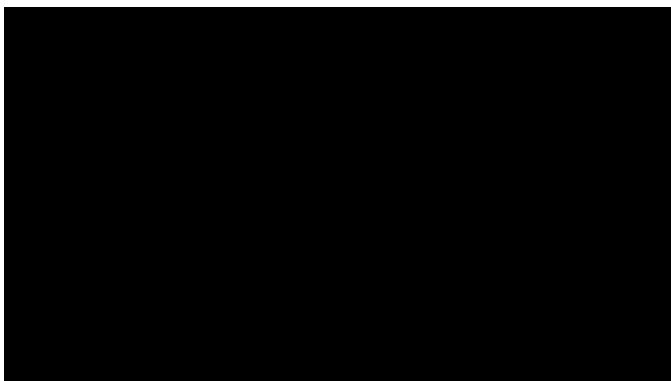
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Section conclusion

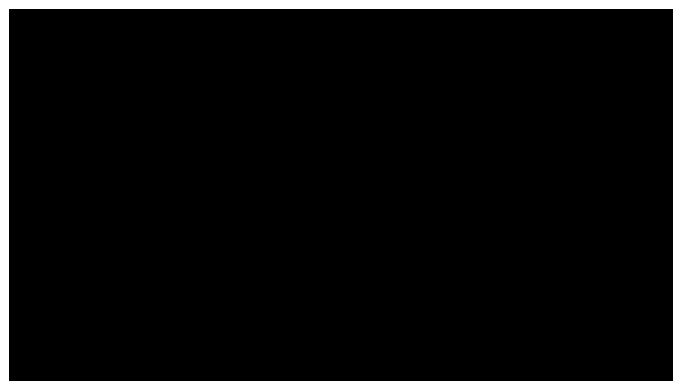
Demonstrate an understanding of business analyst (BA) roles and responsibilities

727

Section overview

Determine how to conduct stakeholder communication

728



729

Stakeholder communication

730

Stakeholder communication

- Informed about the project
- Engaged with the project
- Aligned with the project goals

731

How to conduct stakeholder communication

- Identify stakeholders
- Develop a communication plan
- Communicate regularly
- Use multiple channels
- Be responsive
- Be open to feedback

732

Identify stakeholders

- Determine your stakeholders
- What are their interests?
- What are their concerns?
- Tailor your communication to stakeholder needs
- Ensure that you are engaging all relevant parties

733

Develop a communication plan

- How you will communicate with stakeholders
- What information you will share
- How often you will communicate
- Frequency of communication
- Email
- Meetings
- Project updates

734

Communicate regularly

- Timely
- Transparently
- Helps keep stakeholders informed about project progress
- Allows stakeholders to provide feedback
- Gives stakeholders the opportunity to ask questions

735

Use multiple channels

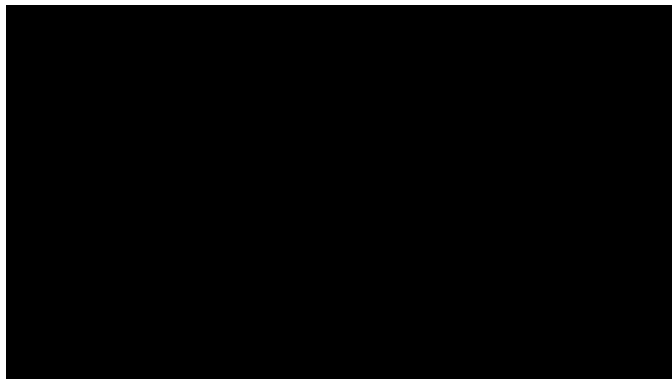
- Stakeholders may prefer different channels of communication
- It is important to use a variety of channels to ensure that you are reaching everyone
- In-person meetings
- Email
- Phone calls
- Video conferencing

736

Be open to feedback

- Valuable
- Can help to improve projects
- Feedback
- Concerns

737



738

Appropriate stakeholder communication channels and tools

739

Appropriate stakeholder communication channels and tools

- Stakeholder needs
- Stakeholder preferences
- Nature of the information being shared

740

Types of stakeholder communication channels and tools

- Reporting
- Presentation
- Video conferencing
- Project management platform
- Email

741

Reporting

- Used for regular project progress updates
- Dashboards
- Email
- Project management platform
- Presentation tool (PowerPoint, Google slides)

742

Presentations

- Project plans
- Budgets
- Live presentation tool
- Recorded presentation

743

Video conferencing

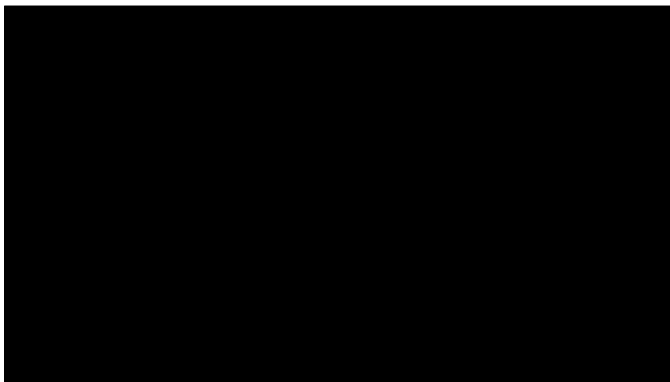
- Used for interactive communication
- To see and hear in real-time
- Interact in discussion and feedback
- Zoom
- Google meet

744

Project management platform

- Used for complex projects with multiple stakeholders
- Asana
- Trello
- Tasks and deadlines
- Communications
- Progress in real-time
- Collaborate with the project team

745



746

Why communication is important for a business analyst between various teams

747

Gathering information

- Business needs
- Project requirements
- Gathering information about features and requirements
- Talking to stakeholders
- Researching industry trends
- Working with subject matter experts

748

Coordinating with different teams

- Development teams
- Testing teams
- Operations teams
- Everyone is aligned
- Everyone is working towards the same goals

749

Clarify and refine requirements

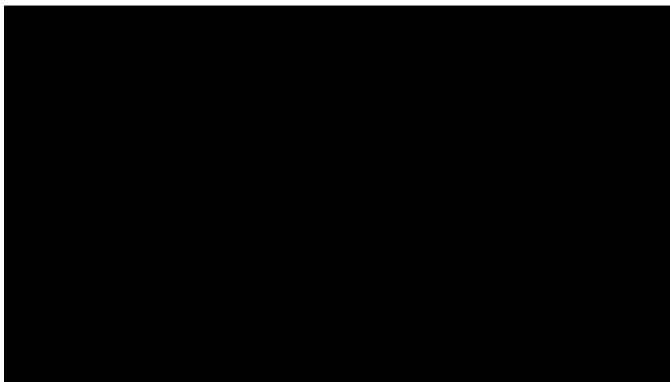
- Work with stakeholders
- Understand stakeholder needs
- Communicate stakeholder needs to the development team

750

Effective communication

- Aids in avoiding misunderstandings
- Goals
- Requirements
- Timelines

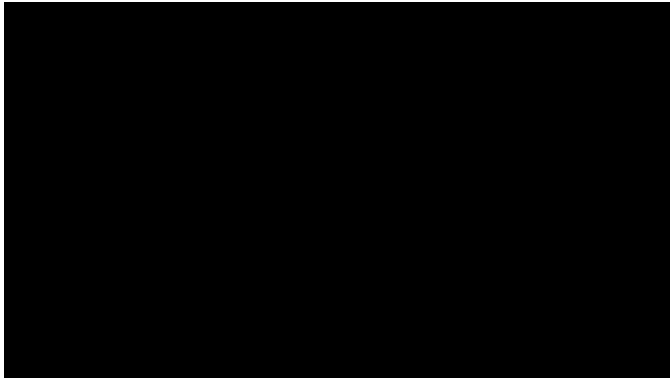
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CAPM Coach

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Section conclusion

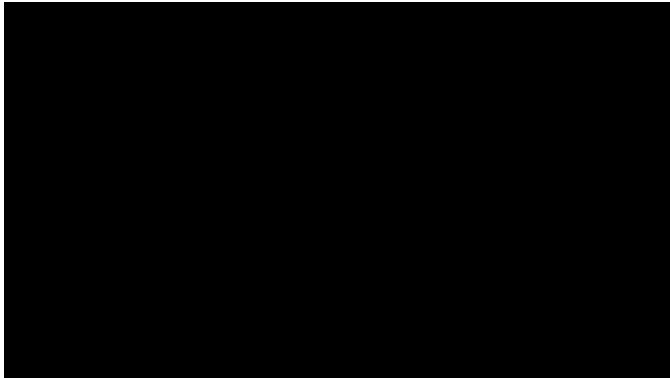
Determine how to conduct stakeholder communication

755

Section overview

Determine how to gather requirements

756



757

How to gather requirements

758

Identify stakeholders

- Helps to understand their needs
- Ensures you are gathering requirements from all relevant parties
- Who the stakeholders are
- Interests
- Concerns

759

Define the project scope

- Ensures that you are gathering the right requirements
- Aids in not overloading the project with unnecessary tasks
- What will be included
- What will be excluded

760

Conduct stakeholder interviews

- Discuss their needs
- Gather their input on the project
- One-on-one interviews
- Focus groups
- Surveys

761

Document the requirements

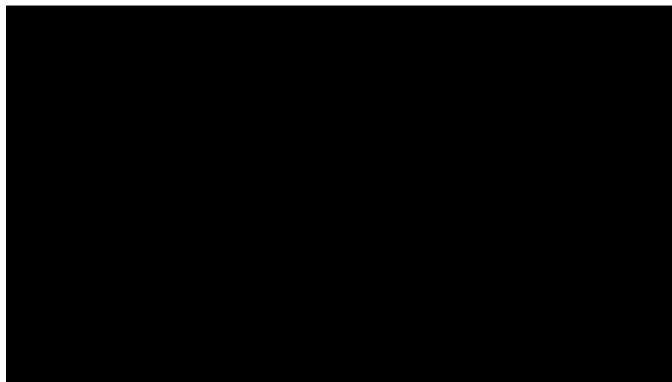
- Use a clear and organized manner
- Details on what the project should achieve
- Constraints
- Limitations
- Relevant deadlines or milestones

762

Review and revise the requirements

- Make necessary revisions based on feedback from stakeholders
- Accurate
- Complete

763



764

Match tools to scenarios

765

User stories

- Short, narrative-style descriptions of a user's needs and goals
- Capture functional requirements of a project from the perspective of the end-user
- Often written in the format:
 - "As a [user], I want to [do something], so that I can [achieve a goal]"

766

Use cases

- Descriptions of how a system or process should behave:
 - Actions
 - Events
- Functional requirements:
 - Detail
 - Structure
 - Detailed description
- Used in software development

767

Interviews

- Interactive
- In-depth
- In person
- Over the phone
- Via video conferencing
- Stakeholders
- Subject matter experts
- Other relevant parties

768

Surveys

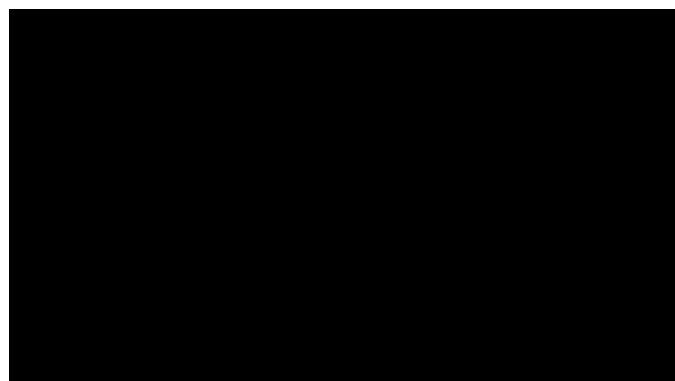
- Anonymous
- Ideal for large number of stakeholders
- Online
- By mail
- Through other means (apps, in-person)

769

Using the Delphi Technique

- Anonymous method
- Survey results analyzed by a third party
- Several rounds of anonymous discussion
- Goal is to gain consensus
- Fear of retribution

770



771

Requirements gathering approach for a situation

772

Requirements gathering approach for a situation

- Stakeholder interviews
- Surveys
- Workshops
- Lessons learned
- Prototyping
- Documentation review
- Observations

773

Stakeholder interviews

- Insights
- Perspectives
- Requirements

774

Workshops

- Used to gather requirements from a group of stakeholders in a collaborative setting
- Brainstorming
- Problem-solving
- Group discussions

775

Prototyping

- Gather feedback
- Gather requirements
- Validate assumptions
- Test ideas early in the development process

776

Documentation review

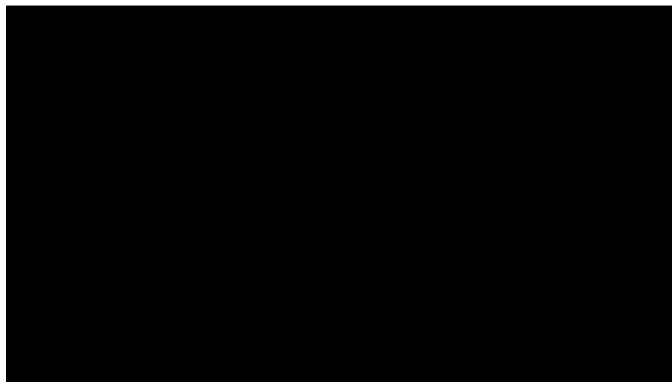
- Business plans
- Policies
- Procedures

777

Observations

- Observing stakeholders can provide valuable insights
- Passive observation
- Active observation
- Shadowing

778



779

Requirements traceability matrix

780

Requirements traceability matrix

- Traces the relationships between requirements and the products or deliverables that they are intended to support
- Lists requirements for a project
- Shows how requirements are related
- Displays overall project goals

781

Requirements traceability matrix

- Requirement type (functional or non-functional)
- Stakeholder group (user or business requirements)
- Requirement status
- Who is responsible for implementation
- Dependencies
- Constraints

782

Requirements traceability matrix

#	A	B	C	D	E	F	G	H	I	J	K	L	M	
1	REQUIREMENTS TRACEABILITY MATRIX						REQUIREMENTS TRACEABILITY MATRIX							
2	Project Name		Project Name		Project Name		Project Name		Project Name		Project Name		Project Name	
3	Release Center		Release Center		Release Center		Release Center		Release Center		Release Center		Release Center	
4	Project Manager		Project Manager		Project Manager		Project Manager		Project Manager		Project Manager		Project Manager	
5	Project Description		Project Description		Project Description		Project Description		Project Description		Project Description		Project Description	
6	Req ID	Technical Assumptions and/or Customer Needs	Functional Requirement	Status	Architecture/Design Document	Technical Specification	System Component	Software Module	Test Case Number	Valid To	Implemented In	Verification		
7	101													
8	102													
9	103													
10	104													
11	105													

783

Product backlog

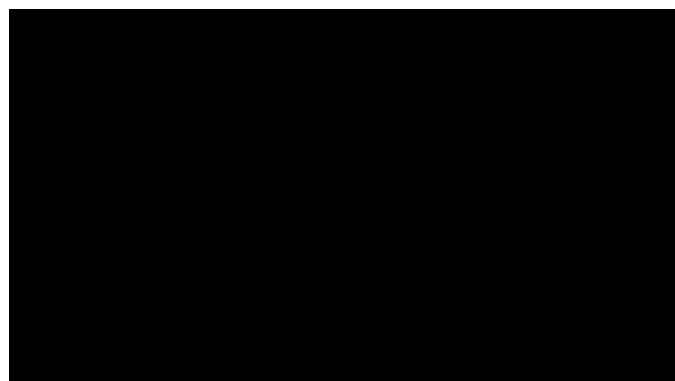
- Product
- Solution
- A prioritized list of work to be done
- Most important items at the top
- Typically used in agile/scrum

784

Product backlog

- Focused on the implementation of requirements rather than traceability
- A living document
- Updated
- Refined
- Prioritize work

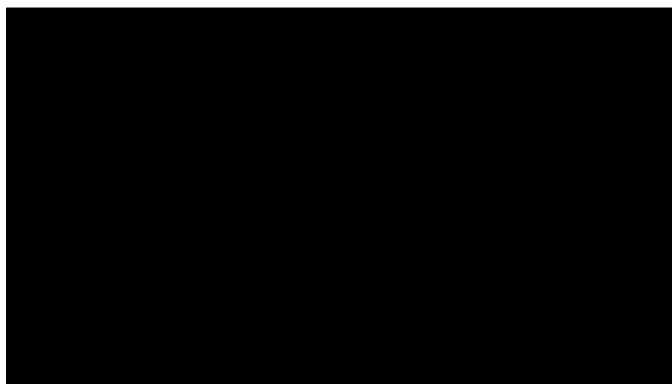
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CAPM Coach

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Section conclusion

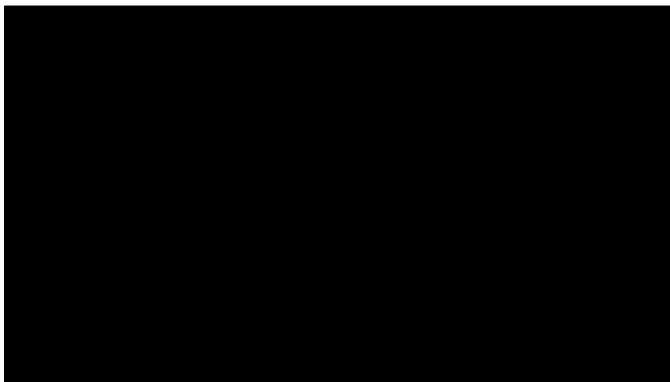
Determine how to gather requirements

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Section overview

Demonstrate an understanding of product roadmaps

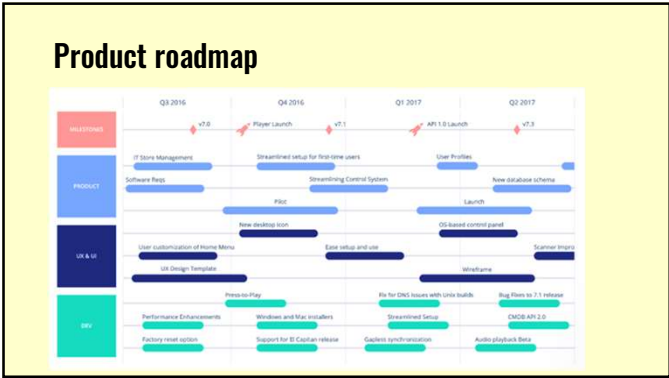
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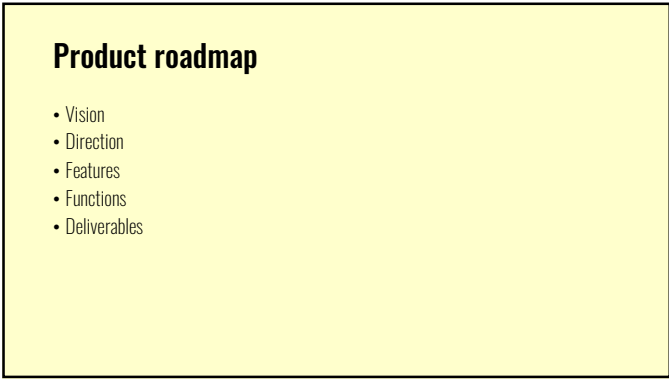
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Product roadmap

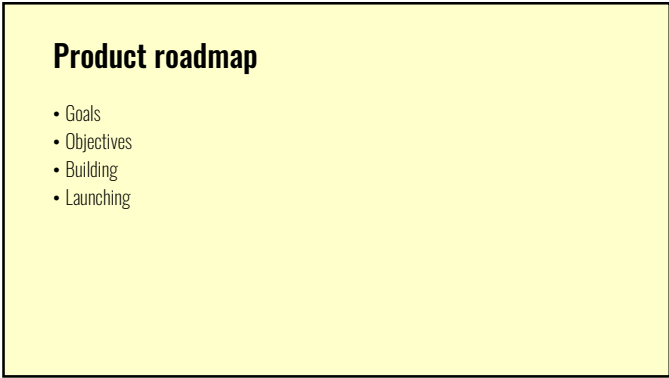
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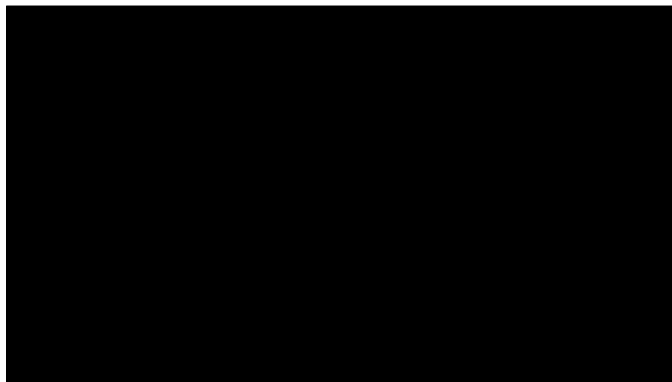


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Product roadmap

- What the product team is working on
- When certain features can be expected
- When updates will be released

796



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Application of a product roadmap

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Application of a product roadmap

- Setting goals and objectives
- Prioritizing features and functions
- Aligning efforts and resources
- Communicating with stakeholders
- Tracking progress

799

Setting goals and objectives

- Product vision and direction
- Specific goals and objectives
- Increasing user engagement
- Improving the user experience
- Expanding the product's reach to new markets

800

Prioritizing features and functions

- Importance
- Impact
- Focusing on the most valuable work
- Not trying to do too much at once

801

Aligning efforts and resources

- Coordinate the efforts of different teams
- Ensure that everyone is working towards the same goals
- Allocate resources effectively

802

Communicating with stakeholders

- An understanding of what the product team is working on
- When they can expect certain features
- When updates will be released
- Builds trust and credibility
- Keeps everyone informed about progress

803

Tracking progress

- Identify any potential roadblocks
- Identify issues that may need to be addressed
- Keep the team on track
- Deliver the product on time

804



805

Which components go to which releases

806

Which components go to which releases

- Priority
- Dependencies
- Feasibility
- Market demand
- Resource constraints

807

Priority

- The most important components or features will typically be prioritized for earlier releases
- Less important or lower-impact items may be delayed until later releases
- Eat your dessert first

808

Dependencies

- Some components may depend on the completion of other components
- Example: a new feature may rely on an underlying infrastructure change
- Mandatory dependency
- Hard logic

809

Feasibility

- Teams may need to consider whether certain components are technically feasible
- If a component would require significant additional effort or resources to implement, it may be better suited for a future release

810

Market demand

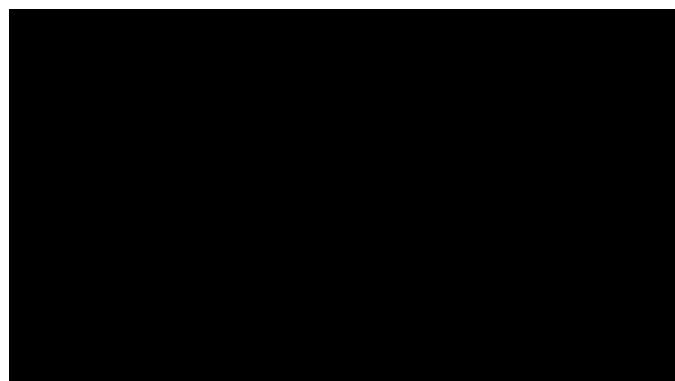
- Consider market demand when determining which components to include in a release
- Example: if there is a high level of demand for a particular feature, the team may prioritize its inclusion in an upcoming release

811

Resource constraints

- Budget
- Personnel
- Time
- Availability

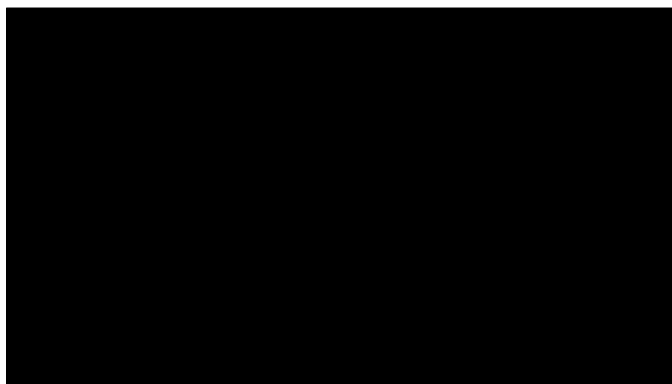
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CAPM Coach

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815

Section conclusion

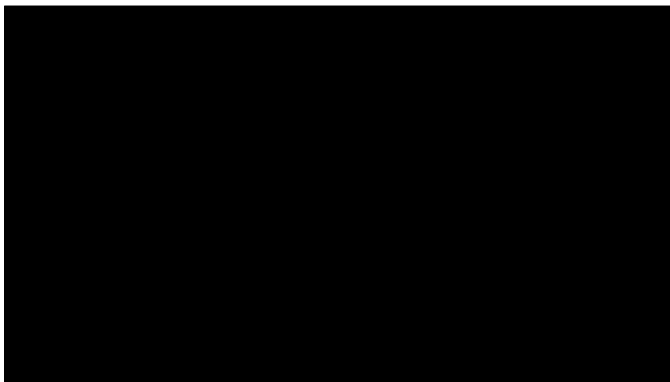
Demonstrate an understanding of product roadmaps

816

Section overview

Determine how project methodologies influence business analysis processes

817



818

Project methodologies influence on business analysis processes

819

Project methodologies influence on business analysis processes

- Waterfall
- Agile
- Scrum
- Lean
- Kanban

820

Waterfall

- Requirements are defined upfront
- Detailed plans and designs
- Implementation of solutions

821

Agile

- Defined
- Refined
- Iterative
- Incremental

822

Scrum

- Small
- Incremental
- Regularly reassessed
- Adjusted
- Changes are based on feedback from stakeholders

823

Lean

- Maximizing value
- Minimizing waste
- Identified
- Eliminated

824

Kanban

- Visualizing the flow of work
- Continuously improving the process to increase efficiency

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826

The role of a business analyst in adaptive approaches

827

The role of a business analyst in adaptive approaches

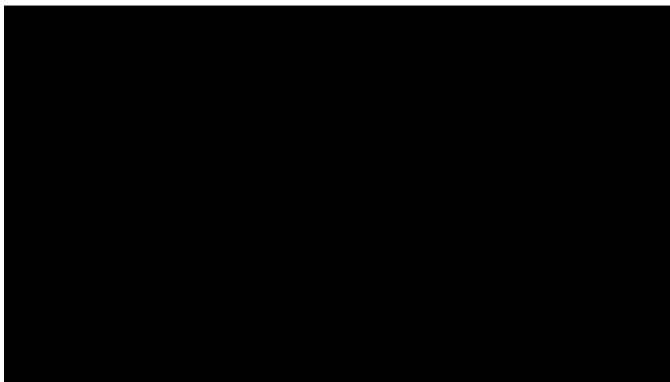
- Identify project work
- Prioritize value
- Helps plan for delivering work
- Identifying risk and issues

828

The role of a business analyst in predictive plan-based approaches

- Define project scope
- Create a detailed project plan
- Risks
- Issues
- Stays on track
- Meets defined objectives

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CAPM Coach

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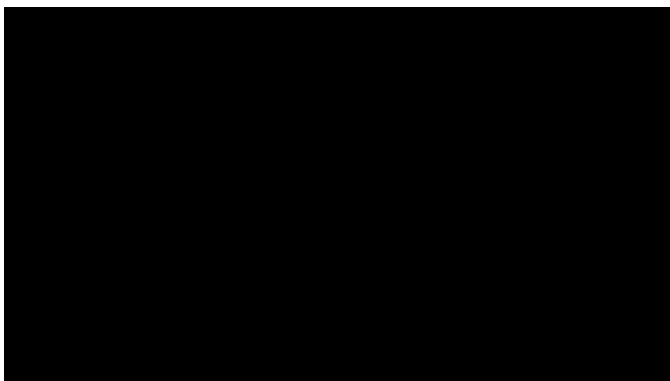


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Section conclusion

Determine how project methodologies influence business analysis processes

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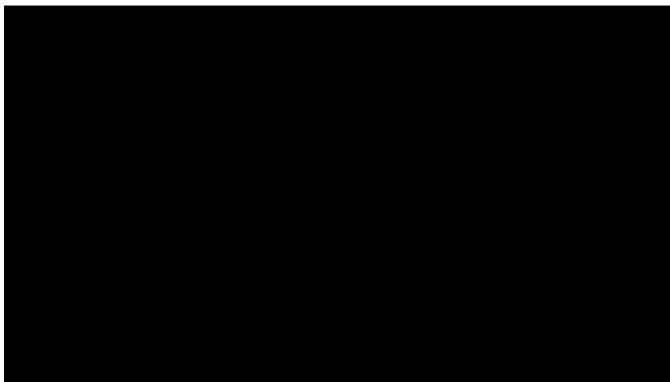


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Section overview

Validate requirements through product delivery

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836

Validate requirements through product delivery

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Testing

- Requirements can be validated through testing
- Actual product behavior
- Expected behavior
- Manual testing methods
- Automated testing methods

838

User acceptance testing (UAT)

- Needs
- Expectations
- The product is tested by stakeholders

839

Review and acceptance

- Review requirements
- Provide feedback
- Completeness
- Correctness
- Clarity

840

Demonstration

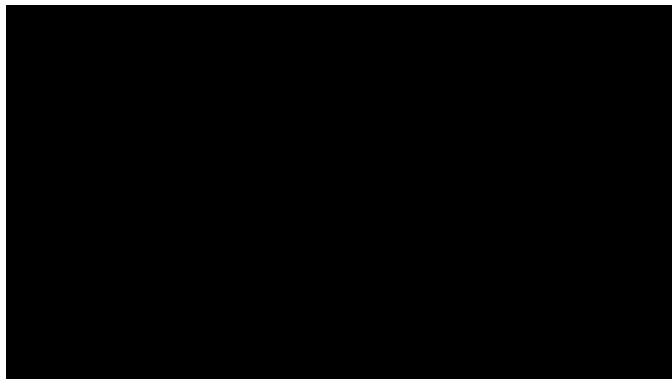
- Showing the product to stakeholders and receiving feedback
- Expectations
- Requirements
- Needs

841

Prototyping

- Feasibility prototypes
 - Just enough code to see if it's feasible
 - Understand technical risk
- Low-fidelity user prototypes
 - An interactive wireframe
 - Test the workflow
 - Simulates process to identify usability
- High-fidelity user prototypes
 - Working simulation
 - Communicating proposed solution
- Live-data prototypes
 - Limited implementation to prove solution
 - Real data / real traffic

842



843

Acceptance criteria (the action of defining changes based on the situation)

844

Acceptance criteria (the action of defining changes based on the situation)

- Customers
- End users
- Complete
- Ready for acceptance
- Behaviors
- Features
- Characteristics

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Acceptance criteria

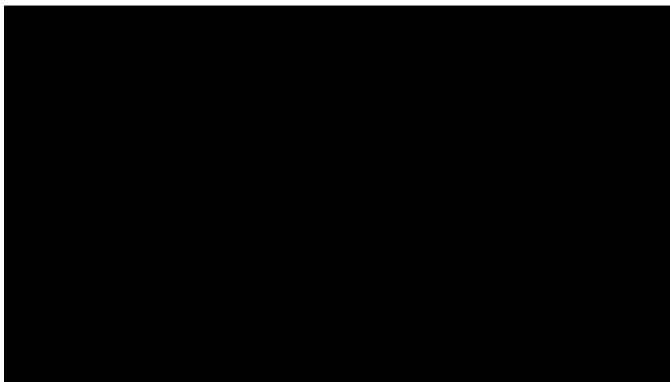
- Typically defined during the requirements gathering phase
- Clear, specific, measurable
- Based on the needs and expectations of the end user
- Aligned with the overall goals and objectives of the project

846

Examples of acceptance criteria

- Specific feature or functionality
- Handle a certain number of users or transactions
- Able to process data within a certain time frame
- Certain level of performance or reliability
- Must meet accessibility or security standards

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CAPM Coach

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850

Section conclusion

Validate requirements through product delivery

851

CAPM exam prep seminar

Course conclusion

852

CAPM requirements

- Education requirements
 - High school diploma
 - 23 contact hours needed
- Certification maintenance requirements
 - 15 PDUs per three-year cycle to maintain certification
- Exam information
 - 3 hours, 150 questions
 - 15 pre-test questions

853

Demonstrate an understanding of the various project life cycles and processes

854

Demonstrate an understanding of project management planning

855

Demonstrate an understanding of project roles and responsibilities

856

Determine how to follow and execute planned strategies or frameworks

857

Demonstrate an understanding of common problem-solving tools and techniques

858

Explain when it is appropriate to use a predictive, plan-based approach

859

Demonstrate an understanding of a project management plan schedule

860

Determine how to document project controls of predictive, plan-based projects

861

Explain when it is appropriate to use an adaptive approach

862

Determine how to plan project iterations

863

Determine how to document project controls for an adaptive project

864

Explain the components of an adaptive plan

865

Determine how to prepare and execute task management steps

866

Demonstrate an understanding of business analyst (BA) roles and responsibilities

867

Determine how to conduct stakeholder communication

868

Determine how to gather requirements

869

Demonstrate an understanding of product roadmaps

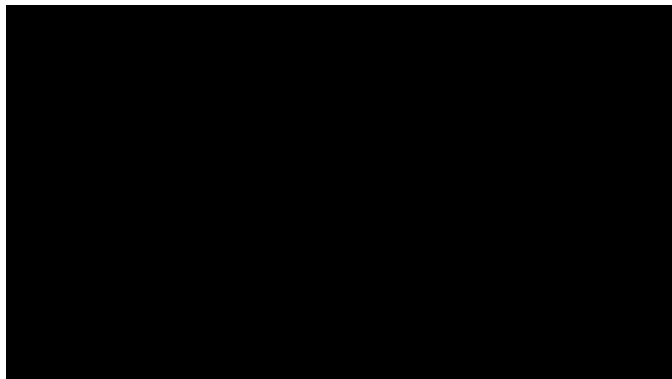
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Determine how project methodologies influence business analysis processes

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Validate requirements through product delivery

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